eddycovariance_HC

AUTHOR
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Gas exchange across agriculture management strategy

This study examines the differences in average and cumulative greenhouse gas exchange from two different fields located in the Hudson Valley (Columbia County). Both fields are grain fields growing primarily corn and soy, with occasional cover cropping and rotations of barley. The main difference in farming strategy is organic certification, with one field being organic and the other conventionally-operated. However they share many practices, including tillage, fertilizing, cover cropping. However they each have stochastic management events not shared by the other, including herbicide use (conventional) and growing alfalfa hay (organic).

Data were collected from 2018-2022 at two eddy covariance towers equipped with carbon dioxide sensors, one located in each field. Between 2019 and 2020, data were also collected using handheld survey gas exchange units (for both soil respiration and NEP). (Survey data were collected in more years, but these data are not yet cleaned and analyzable.) Other data sources, such as photosynthetically active radiation (PAR); management records; and soil organic matter (SOM) were also collected for subsets of the period during which the towers were deployed and collecting data.

The purposes of these analyses include demonstrating the difference in average and cumulative gas exchange across treatments (organic vs. conventional) using eddy covariance data, augmented and supported by survey data and meta-data where necessary and/or appropriate.

Libraries

Upload cleaned data to global environment

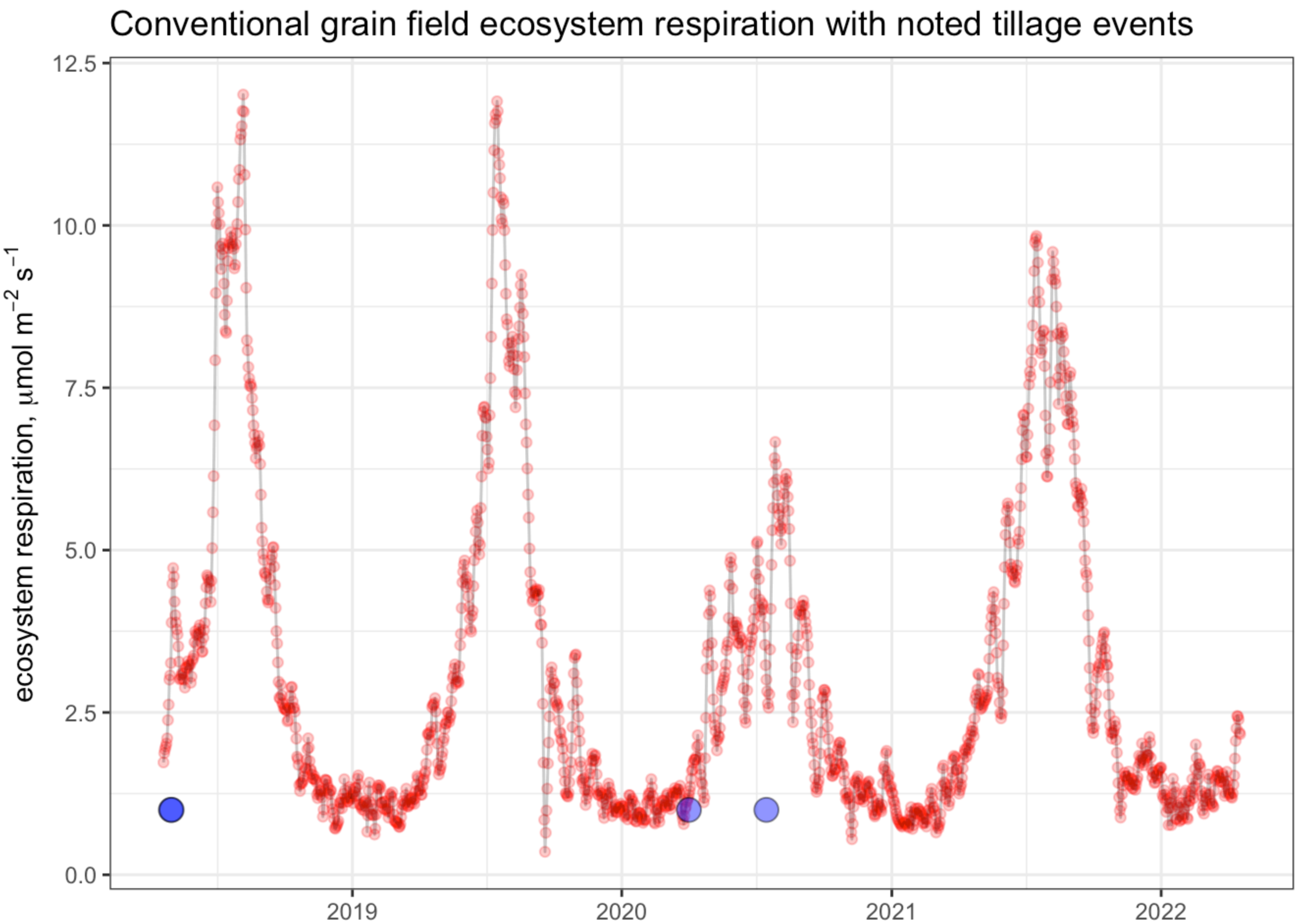
Eddy covariance, survey, PAR/weather at each tower, and management data:

Add management data to the EC, chamber data:

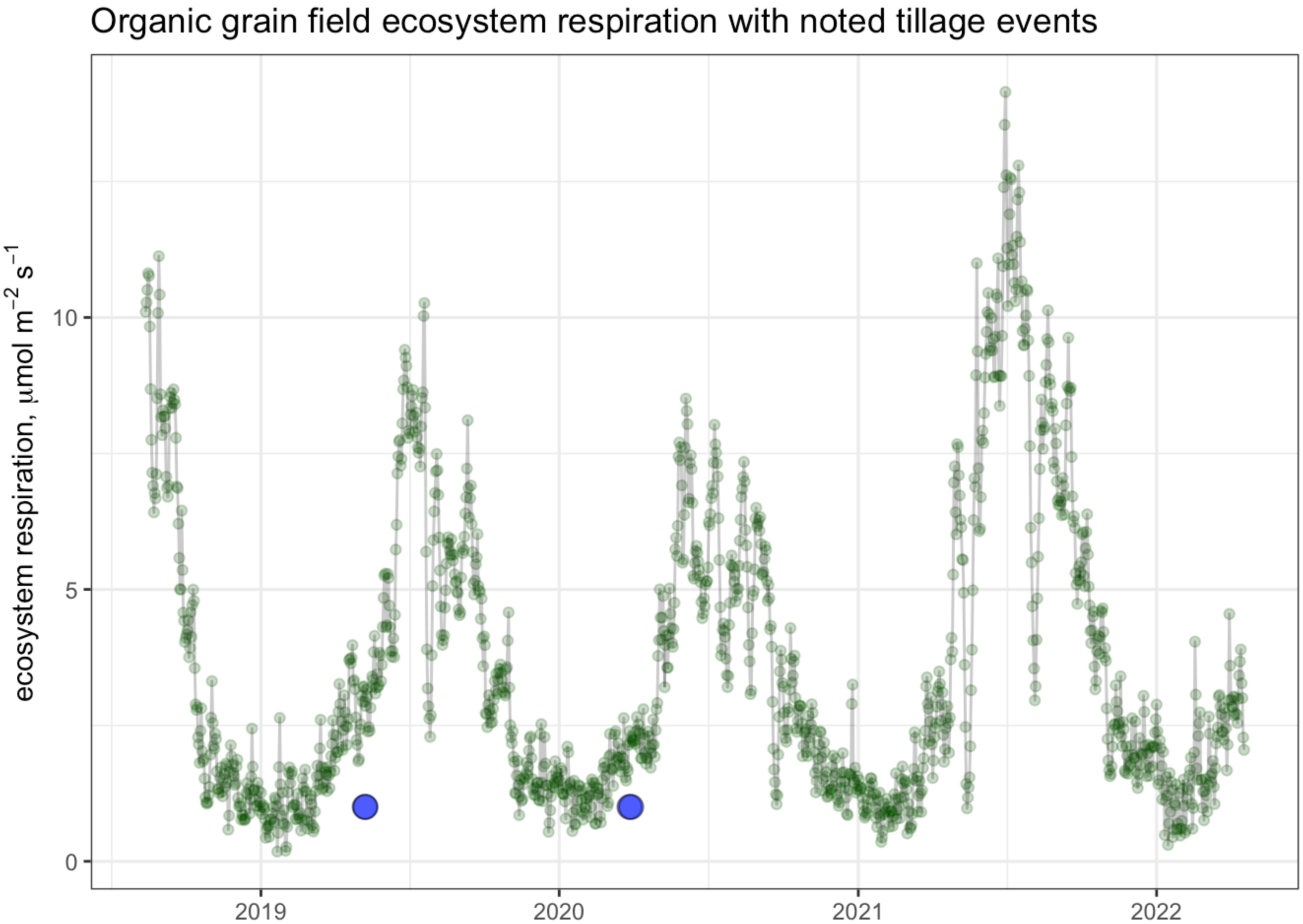
Using a custom function, calculate several continuous variables for each management action. The custom function is called “merge_management_data_timeseries.r” and is informed by regionally-specific characteristics like the expected crop-specific phenology plus days since management action.

I wrote this initial ec_daily_management df to .csv and made some manual changes to it, based on the management dataset and a line-by-line read to make sure the new continuous and categorical variables accurately reflect it. (see process notes document.) Most of these were done in the CSV file itself. Below, I am uploading the changed document. This involved doing some logic-based assignments of crop stage for the overwintering winter rye cover crop, and the one case of summer rye that overwintered on Kukon in 2020/2021.

Plot conventional Reco with tillage events:



Plot organic Reco with tillage events



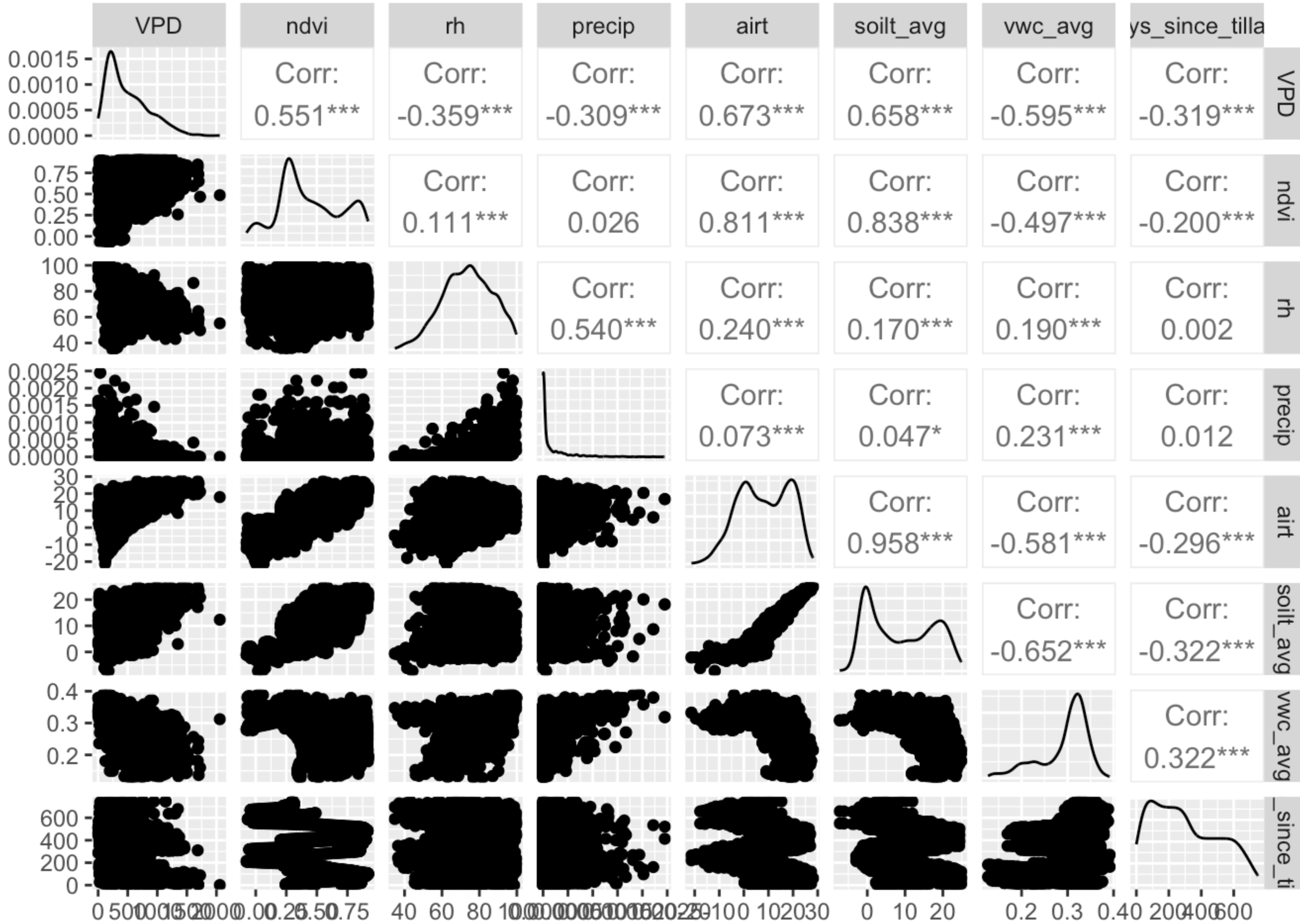
Analyses

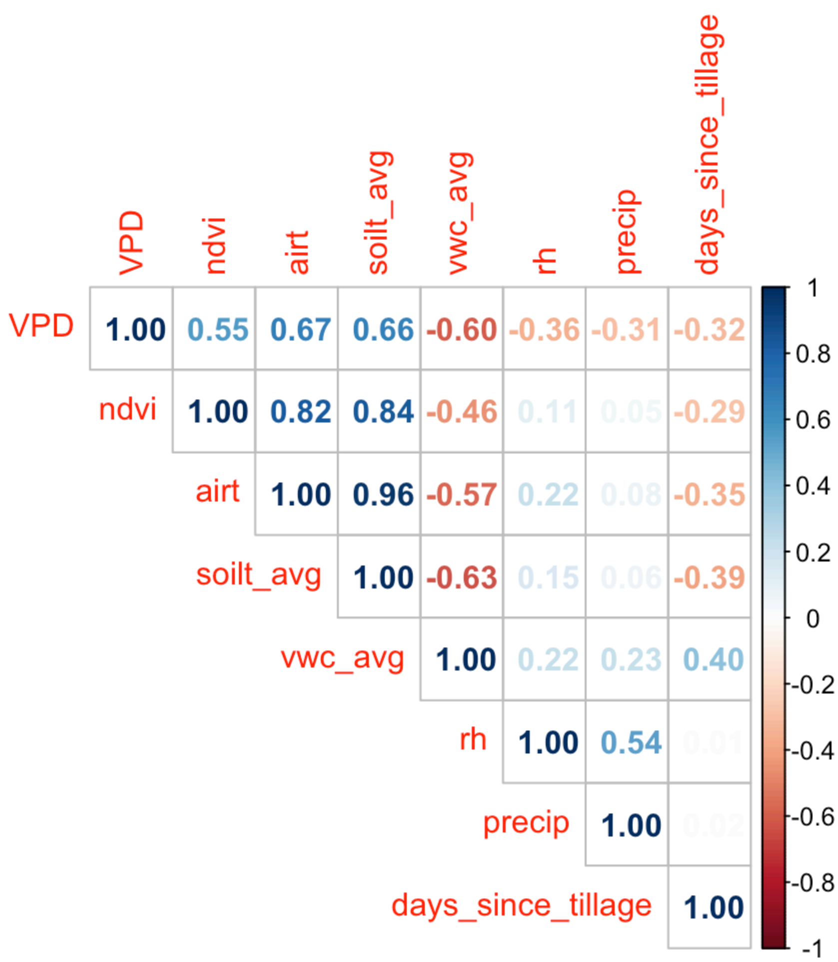
GAM: EC data, daily averages

Assessing daily average trends is useful because it gives mechanistic insight into how gas exchange varies across management types. This analysis tells us when and how management moderates fluxes throughout the seasons we measured.

NEE

GAM model development: daily average


VPD
ndvi
rh
precip
airt
soilt_avg
vwc_avg
ys_since_tilla
Corr: 0.551***
Corr: -0.359***
Corr: -0.309***
Corr: 0.673***
Corr: 0.658***
Corr: -0.595***
Corr: -0.319***
Corr: 0.111***
Corr: 0.026
Corr: 0.811***
Corr: 0.838***
Corr: -0.497***
Corr: -0.200***
Corr: 0.540***
Corr: 0.240***
Corr: 0.170***
Corr: 0.190***
Corr: 0.002
Corr: 0.073***
Corr: 0.047*
Corr: 0.231***
Corr: 0.012
Corr: 0.958***
Corr: -0.581***
Corr: -0.296***
Corr: -0.652***
Corr: -0.322***
Corr: 0.322***

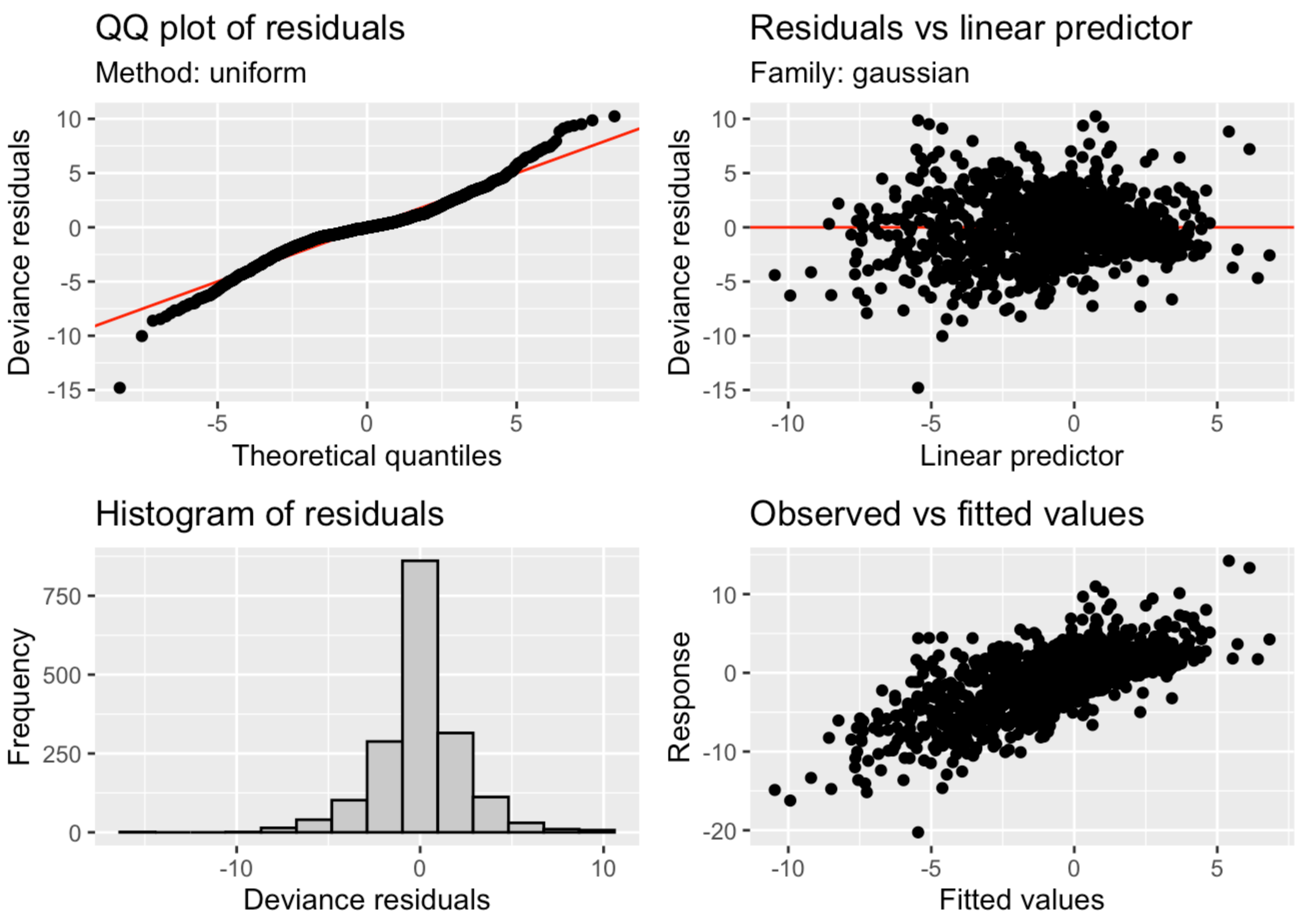

VPD
ndvi
airt
soilt_avg
vwc_avg
rh
precip
days_since_tillage
VPD 1.00 0.55 0.67 0.66 -0.60 -0.36 -0.31 -0.32
ndvi 1.00 0.82 0.84 -0.46 0.11 0.05 -0.29
airt 1.00 0.96 -0.57 0.22 0.08 -0.35
soilt_avg 1.00 -0.63 0.15 0.06 -0.39
vwc_avg 1.00 0.22 0.23 0.40
rh 1.00 0.54 0.01
precip 1.00 0.02
days_since_tillage 1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

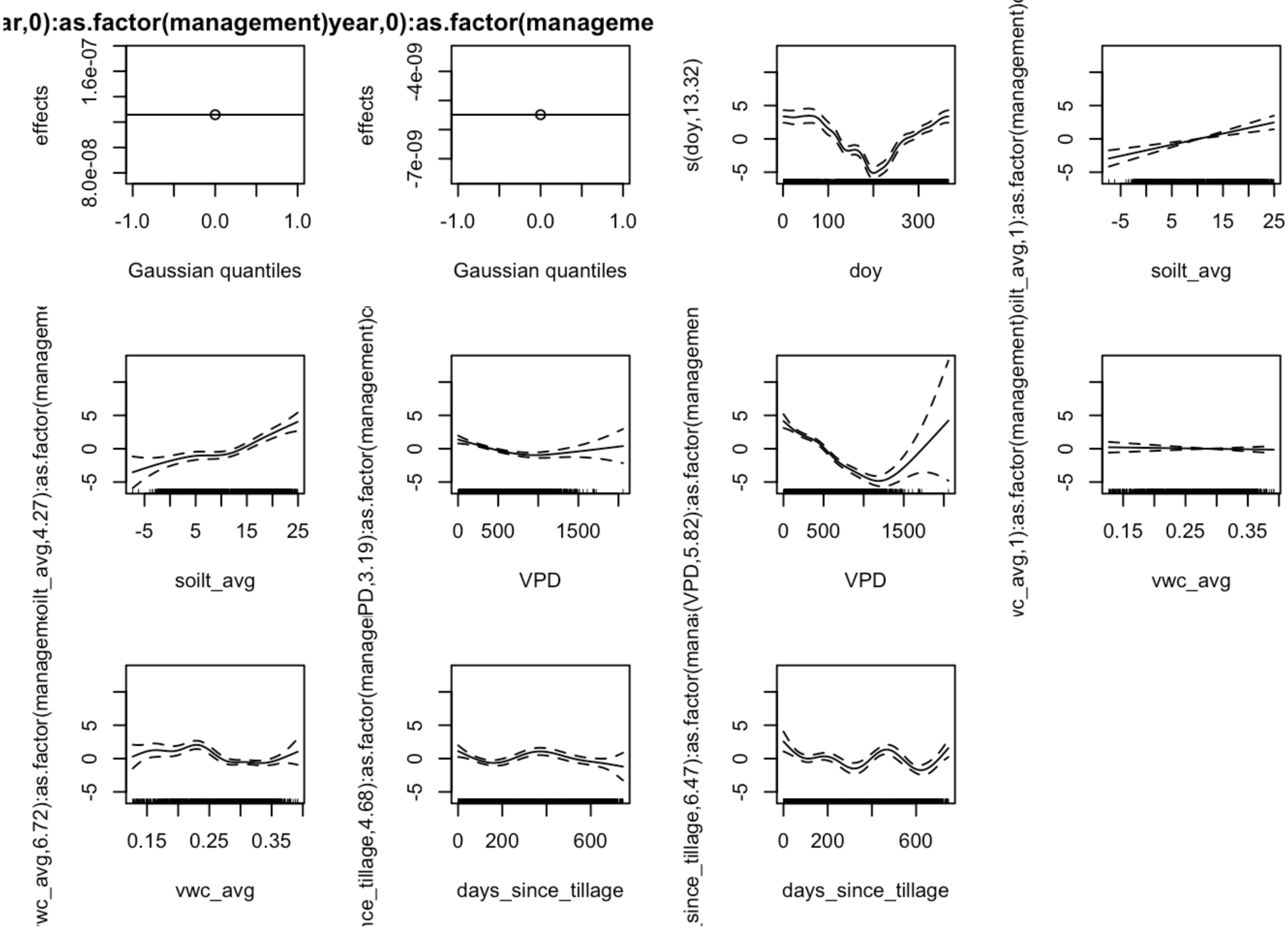
Model testing, average NEE over time: simple with main effects, then with different responses by management, then with replacement of NDVI with crop stage to account for phenology. resource on GAMs: https://deepwiki.com/cran/mgcv/2.1-the-gam()-function resource on smoothing and interaction variables: https://deepwiki.com/cran/mgcv/3.1-types-of-smooths

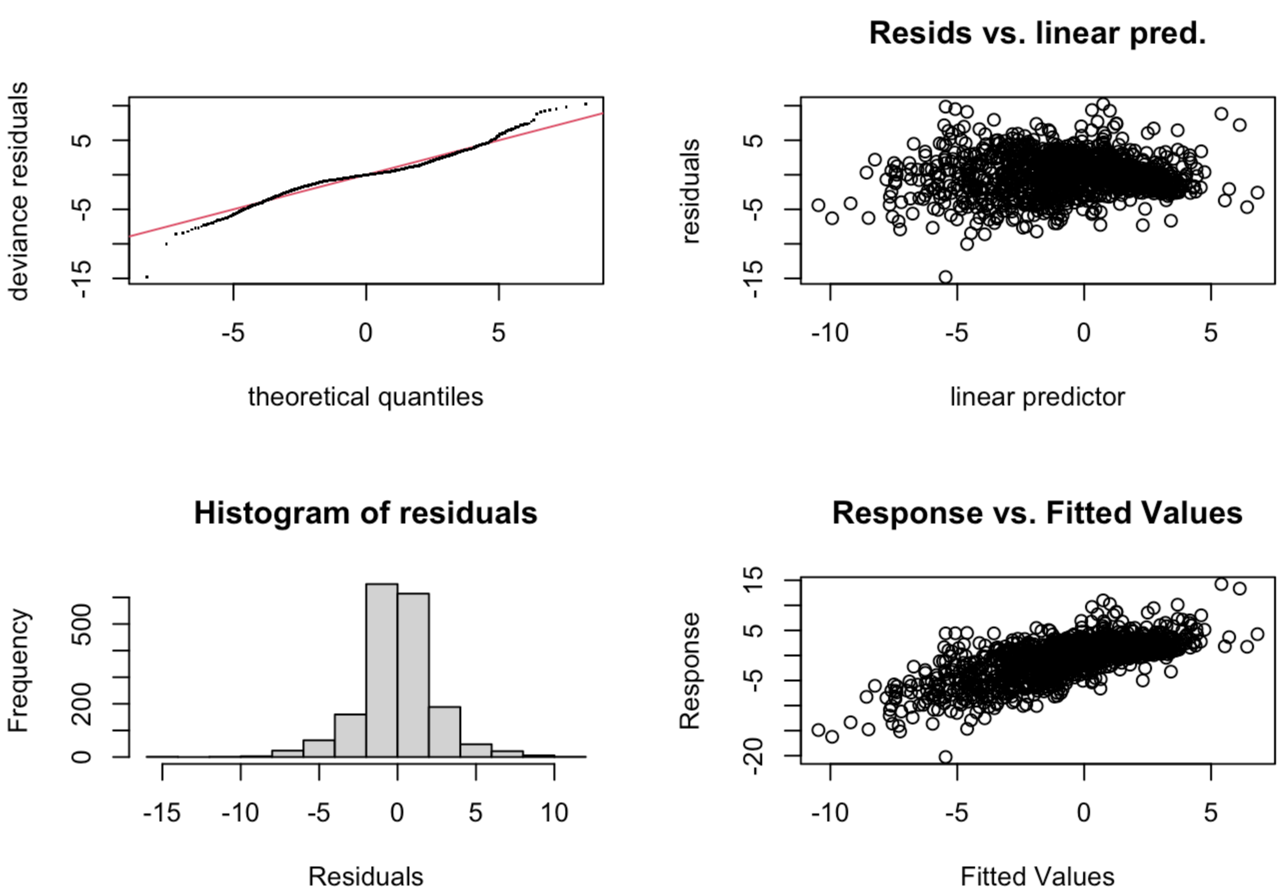
```
           df      AIC
g3   60.75865 8302.817
g3.2 81.75316 8291.385
g3.3 91.15208 8271.081

Analysis of Deviance Table

Model 1: NEE ~ management + crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + s(soilt_avg, by = as.factor(management),
    k = 10) + s(VPD, by = as.factor(management), k = 10) + s(vwc_avg,
    by = as.factor(management), k = 10) + s(days_since_tillage,
    by = as.factor(management), k = 8)
Model 2: NEE ~ management + crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
  Resid. Df Resid. Dev     Df Deviance Pr(>Chi)
1    1713.4      10308
2    1687.9      10004 25.525    304.6 0.001436 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Deviance Table

Model 1: NEE ~ management + crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + s(soilt_avg, by = as.factor(management),
    k = 10) + s(VPD, by = as.factor(management), k = 10) + s(vwc_avg,
    by = as.factor(management), k = 10) + s(days_since_tillage,
    by = as.factor(management), k = 8)
Model 2: NEE ~ management * crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
  Resid. Df Resid. Dev    Df Deviance Pr(>Chi)
1    1713.4    10308.3
2    1676.2     9786.5 37.19   521.85 2.24e-06 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Going to use simple crop phase as a categorical response, rather than NDVI as a smoothing response, to investigate the effects of management on the different life stages of the crops across the time series.




ar,0):as.factor(management)year,0):as.factor(manageme
effects
Gaussian quantiles
s(doy,13.32)
doy
soilt_avg
VPD
vwc_avg
days_since_tillage


deviance residuals
theoretical quantiles
Resids vs. linear pred.
residuals
linear predictor
Histogram of residuals
Frequency
Residuals
Response vs. Fitted Values
Response
Fitted Values

```
Method: REML   Optimizer: outer newton
full convergence after 10 iterations.
Gradient range [-0.00135249,0.0005358389]
(score 4159.653 & scale 5.748221).
eigenvalue range [-3.906107e-07,878.598].
Model rank =  200 / 200

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                                                              k'      edf
s(year):as.factor(management)conventional              1.00e+00 9.52e-08
s(year):as.factor(management)organic                   1.00e+00 2.93e-08
s(doy)                                                 1.80e+01 1.27e+01
ti(soilt_avg,vwc_avg):as.factor(management)conventional 4.90e+01 5.29e+00
ti(soilt_avg,vwc_avg):as.factor(management)organic      4.90e+01 1.21e+01
s(soilt_avg):as.factor(management)conventional         9.00e+00 2.43e+00
s(soilt_avg):as.factor(management)organic              9.00e+00 4.39e+00
s(VPD):as.factor(management)conventional               9.00e+00 3.20e+00
s(VPD):as.factor(management)organic                    9.00e+00 5.98e+00
s(vwc_avg):as.factor(management)conventional           9.00e+00 1.00e+00
s(vwc_avg):as.factor(management)organic                9.00e+00 6.74e+00
s(days_since_tillage):as.factor(management)conventional 7.00e+00 4.90e+00
s(days_since_tillage):as.factor(management)organic      7.00e+00 5.75e+00
                                                        k-index p-value
s(year):as.factor(management)conventional                  0.49  <2e-16 ***
s(year):as.factor(management)organic                       0.49  <2e-16 ***
s(doy)                                                     1.14    1.00
ti(soilt_avg,vwc_avg):as.factor(management)conventional    1.03    0.91
ti(soilt_avg,vwc_avg):as.factor(management)organic         1.03    0.91
s(soilt_avg):as.factor(management)conventional             1.00    0.45
s(soilt_avg):as.factor(management)organic                  1.00    0.46
s(VPD):as.factor(management)conventional                   1.00    0.55
s(VPD):as.factor(management)organic                        1.00    0.55
s(vwc_avg):as.factor(management)conventional               1.00    0.45
s(vwc_avg):as.factor(management)organic                    1.00    0.44
s(days_since_tillage):as.factor(management)conventional    0.97    0.10 .
s(days_since_tillage):as.factor(management)organic         0.97    0.13
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Family: gaussian
Link function: identity

Formula:
NEE ~ management * crop_stage_simple + s(year, bs = "re", by = as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
```

```
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)

Parametric coefficients:
                                                Estimate Std. Error t value
(Intercept)                                      0.15884    0.37663   0.422
managementorganic                               -1.11203    0.55277  -2.012
crop_stage_simpleearly                           0.87636    0.52965   1.655
crop_stage_simplevegetative                      0.09207    0.41619   0.221
crop_stage_simplereproductive                   -0.11353    0.46936  -0.242
crop_stage_simplegrain_fill                     -0.79441    0.42691  -1.861
crop_stage_simpledormant                        -0.22657    0.53827  -0.421
crop_stage_simplefallow                          1.46194    0.44720   3.269
managementorganic:crop_stage_simpleearly         2.24519    0.71961   3.120
managementorganic:crop_stage_simplevegetative    1.04629    0.54534   1.919
managementorganic:crop_stage_simplereproductive  0.04115    0.61937   0.066
managementorganic:crop_stage_simplegrain_fill   -0.87595    0.57117  -1.534
managementorganic:crop_stage_simpledormant      -0.71766    0.76112  -0.943
managementorganic:crop_stage_simplefallow        2.66124    0.93190   2.856
                                                Pr(>|t|)
(Intercept)                                      0.67328
managementorganic                                0.04440 *
crop_stage_simpleearly                           0.09819 .
crop_stage_simplevegetative                      0.82495
crop_stage_simplereproductive                    0.80890
crop_stage_simplegrain_fill                      0.06294 .
crop_stage_simpledormant                         0.67386
crop_stage_simplefallow                          0.00110 **
managementorganic:crop_stage_simpleearly         0.00184 **
managementorganic:crop_stage_simplevegetative    0.05520 .
managementorganic:crop_stage_simplereproductive  0.94703
managementorganic:crop_stage_simplegrain_fill    0.12531
managementorganic:crop_stage_simpledormant       0.34587
managementorganic:crop_stage_simplefallow        0.00435 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                                              edf Ref.df      F
s(year):as.factor(management)conventional              9.521e-08  1.000  0.000
s(year):as.factor(management)organic                  2.926e-08  1.000  0.000
s(doy)                                                 1.266e+01 18.000  9.624
ti(soilt_avg,vwc_avg):as.factor(management)conventional 5.291e+00  7.499  1.976
ti(soilt_avg,vwc_avg):as.factor(management)organic      1.212e+01 16.014  2.154
s(soilt_avg):as.factor(management)conventional         2.427e+00  3.028  4.611
s(soilt_avg):as.factor(management)organic              4.390e+00  5.326 10.328
s(VPD):as.factor(management)conventional               3.205e+00  4.061  9.301
s(VPD):as.factor(management)organic                    5.982e+00  7.080 50.957
s(vwc_avg):as.factor(management)conventional           1.004e+00  1.007  0.228
s(vwc_avg):as.factor(management)organic                6.745e+00  7.843  2.593
s(days_since_tillage):as.factor(management)conventional 4.899e+00  5.733  4.057
```

```
s(days_since_tillage):as.factor(management)organic      5.752e+00  6.461  4.575
                                                          p-value
s(year):as.factor(management)conventional               0.001359 **
s(year):as.factor(management)organic                    0.048225 *
s(doy)                                                   < 2e-16 ***
ti(soilt_avg,vwc_avg):as.factor(management)conventional 0.048708 *
ti(soilt_avg,vwc_avg):as.factor(management)organic      0.004982 **
s(soilt_avg):as.factor(management)conventional          0.003305 **
s(soilt_avg):as.factor(management)organic                < 2e-16 ***
s(VPD):as.factor(management)conventional                 < 2e-16 ***
s(VPD):as.factor(management)organic                      < 2e-16 ***
s(vwc_avg):as.factor(management)conventional            0.633289
s(vwc_avg):as.factor(management)organic                 0.006463 **
s(days_since_tillage):as.factor(management)conventional 0.000689 ***
s(days_since_tillage):as.factor(management)organic      0.000378 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.493   Deviance explained = 51.5%
-REML = 4159.7  Scale est. = 5.7482    n = 1781
```

Interpretation, linear variables:

Note: baseline comparison to conventional management and mature crop stage (estimated NEE = 0.159)

There is a significant effect of organic management on average daily NEE in the mature crop stage, reducing it by 1.1 units when holding other factors constant; this makes sense, in that we've seen greater drawdown of ecosystem CO_2 in the organic field compared to the conventional one when comparing them directly.

There are both trends and significant effects of individual crop stage relative to the baseline estimate (conventional, mature). Compared to the conventional mature, the vegetative stage is slightly higher. Compared to the conventional mature, the grain-fill phase is slightly lower. Compared to the conventional mautre, the fallow phase is significantly higher (1.4 units).

There are also significant interactions between management and crop stage, which are the difference in the crop stage effect between organic and conventional relative to the baseline crop stage; the interaction coefficient therefore tells us how much more or less the difference between that crop stage and the baseline changes under organic compared to conventional management. (To calculate the absolute organic vs. conventional difference in NEE for each crop phase, we'd combine the main effect of organic + the interaction effect of organic on that phase.)

The effect of organic management on the influence of the early crop phase compared to conventional mature is 2.2 units; the effect of organic management on the influence of the vegetative stage compared to conventional mature is 1 unit. The effect of organic management on the influence of the fallow stage compared to conventional mature is 2.7 units.

Interpretation, smoothing variables:

Almost every single smoothing factor is significant in driving NEE, with the exception of VWC (soil moisture) in the conventional field, demonstrating the effect of each crop stage. The F-test and p-values indicate with the smooth term explains a meaningful amount of the variation in the model's ability to predict the data. The EDF (estimated degrees of freedom) indicate the complexity of the smoothing term: close to 1 means close to linear, and larger indicates a very 'wiggly' or nonlinear relationship between the smoothing term and NEE.

Seasonality (doy, day of year) has a significant non-linear effect on NEE, capturing expected seasonal variation. The tensor interaction between soil temperature and moisture indicates a significant non-linear effect of the two as they are combined, the extent to which differs by management – the combined effect is important for NEE, but the estimated complexity of this effect is greater in the organic field than the conventional one. This indicates that there are more nuanced and/or variable effects of this interaction there, perhaps due to more nuanced soil properties, soil microbiome, plant-soil interactions, etc. For soil temp and moisture individually, as well as VPD, the nonlinearity of effect is more pronounced in organic than conventional management, though soil moisture does not significantly moderate NEE in the conventional field at all. Days since tillage are similar.

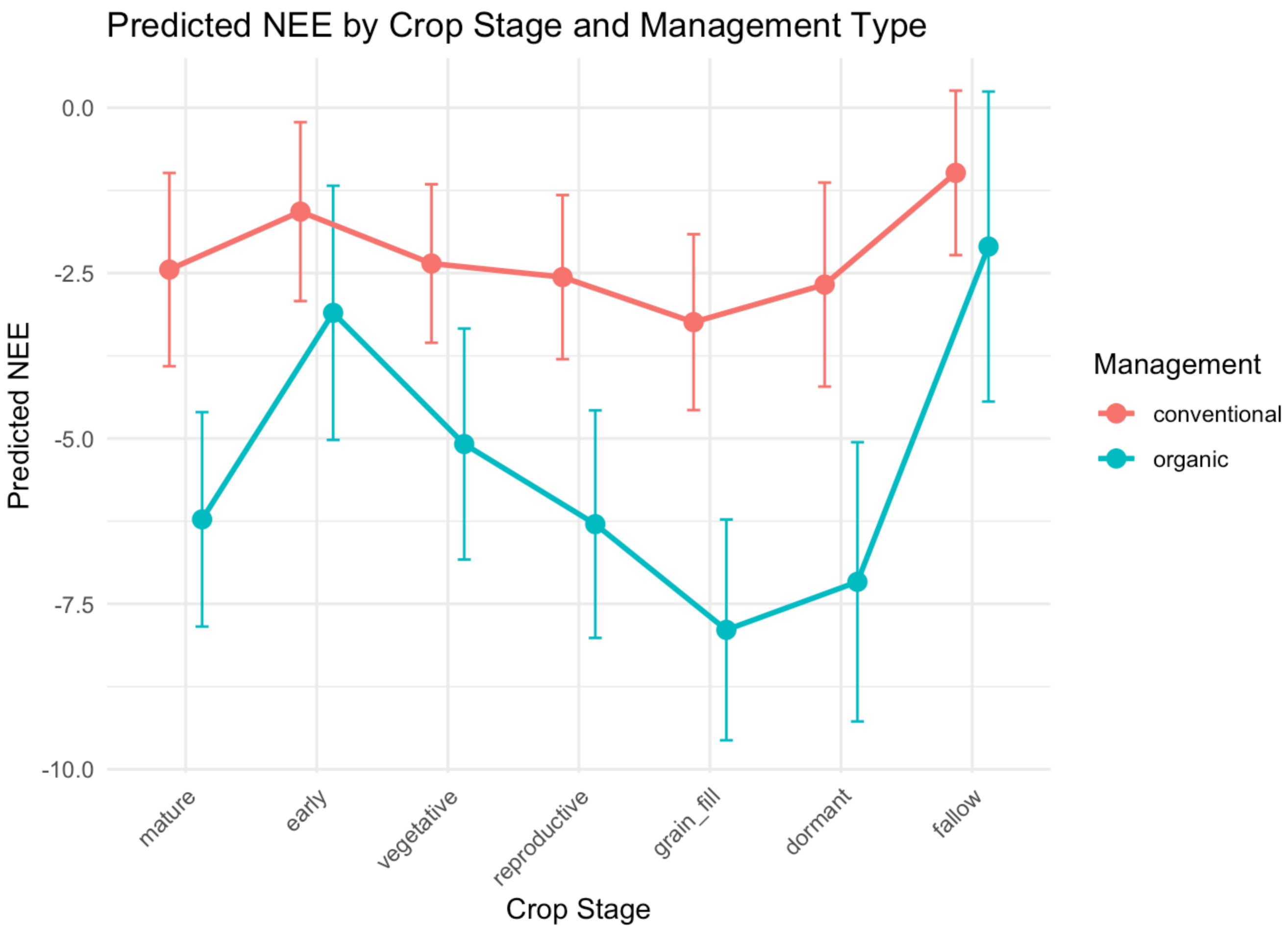
Takeaways:

Organic management creates a tendency toward lower NEE, implying either less CO2 respiration or more uptake via photosynthesis, and/or a combination of the two. Crop stage modifies the effect significantly as well, indicating a different effect of crop stage on NEE depending on management strategy, but overall organic management reduces NEE. The smoothing terms universally indicated nonlinear effects on the organic field's NEE, and almost all except for soil moisture for the conventional field; for the conventional field, the effects were less nonlinear all around.

Interestingly, the random effect of year demonstrates a) that there IS a significant difference in NEE across years, but also that b) these differences are also dependent on management type, indicating that NEE reacts differently to yearly weather patterns, management change, etc., in each field.

Visualize:

Interactions, NEE daily average:



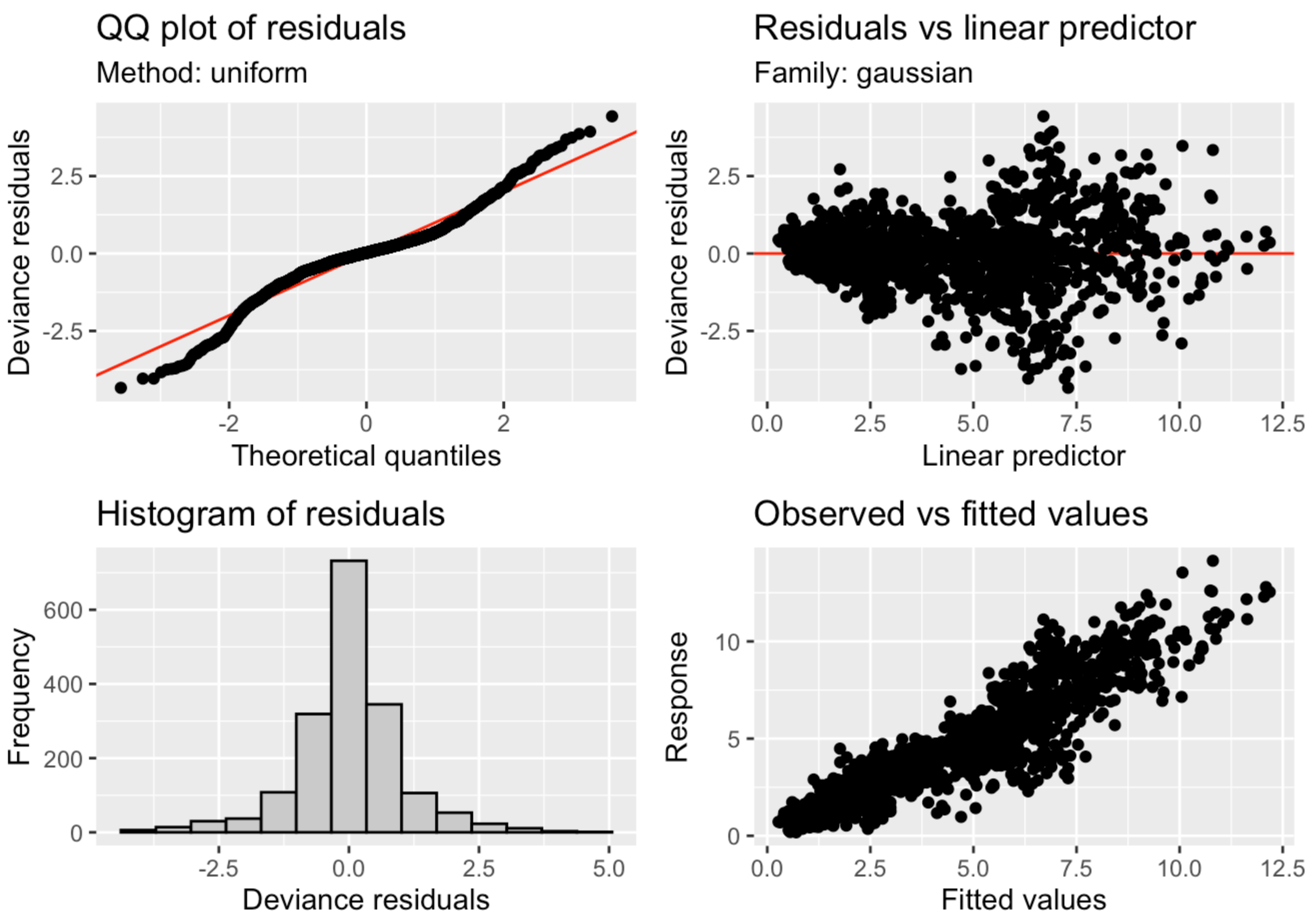
Reco

GAM model development: daily average

```
           df      AIC
r1   70.46897 5430.100
r1.2 92.54523 5392.568
r1.3 98.64308 5312.992

Analysis of Deviance Table

Model 1: Reco ~ management + crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
Model 2: Reco ~ management * crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
```

```
    k = 8)
  Resid. Df Resid. Dev     Df Deviance      F    Pr(>F)
1    1681.9     1926.5
2    1675.8     1830.1 6.1159   96.383 14.658 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assess model performance:




```
Family: gaussian
Link function: identity

Formula:
Reco ~ management * crop_stage_simple + s(year, bs = "re", by = as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)

Parametric coefficients:
                                               Estimate Std. Error t value
(Intercept)                                     3.07036    0.19153  16.031
managementorganic                               1.06870    0.30170   3.542
crop_stage_simpleearly                          0.16742    0.23046   0.726
```

```
crop_stage_simplevegetative                          0.48399    0.18122   2.671
crop_stage_simplereproductive                        0.78167    0.20549   3.804
crop_stage_simplegrain_fill                          1.69384    0.18840   8.991
crop_stage_simpledormant                             0.30100    0.24936   1.207
crop_stage_simplefallow                             -0.26466    0.19526  -1.355
managementorganic:crop_stage_simpleearly             0.44807    0.32352   1.385
managementorganic:crop_stage_simplevegetative        0.05697    0.25181   0.226
managementorganic:crop_stage_simplereproductive -0.42748    0.28176  -1.517
managementorganic:crop_stage_simplegrain_fill   -1.71098    0.25181  -6.795
managementorganic:crop_stage_simpledormant       0.71122    0.36661   1.940
managementorganic:crop_stage_simplefallow        0.67023    0.41364   1.620
                                                Pr(>|t|)    
(Intercept)                                      < 2e-16 ***
managementorganic                               0.000407 ***
crop_stage_simpleearly                          0.467653    
crop_stage_simplevegetative                     0.007639 ** 
crop_stage_simplereproductive                   0.000147 ***
crop_stage_simplegrain_fill                      < 2e-16 ***
crop_stage_simpledormant                        0.227575    
crop_stage_simplefallow                         0.175456    
managementorganic:crop_stage_simpleearly        0.166242    
managementorganic:crop_stage_simplevegetative   0.821033    
managementorganic:crop_stage_simplereproductive 0.129407    
managementorganic:crop_stage_simplegrain_fill    1.5e-11 ***
managementorganic:crop_stage_simpledormant      0.052546 .  
managementorganic:crop_stage_simplefallow       0.105351    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                                           edf Ref.df      F
s(year):as.factor(management)conventional            7.054e-07  1.000  0.000
s(year):as.factor(management)organic                 1.982e-07  1.000  0.000
s(doy)                                               1.424e+01 18.000 12.187
ti(soilt_avg,vwc_avg):as.factor(management)conventional 5.885e+00  7.381  6.105
ti(soilt_avg,vwc_avg):as.factor(management)organic   1.504e+01 19.699  3.098
s(soilt_avg):as.factor(management)conventional       5.557e+00  6.644 46.421
s(soilt_avg):as.factor(management)organic            5.089e+00  6.130 40.295
s(VPD):as.factor(management)conventional             4.560e+00  5.667  2.457
s(VPD):as.factor(management)organic                  2.796e+00  3.551 17.573
s(vwc_avg):as.factor(management)conventional         1.003e+00  1.005 28.373
s(vwc_avg):as.factor(management)organic              6.641e+00  7.713  4.756
s(days_since_tillage):as.factor(management)conventional 4.495e+00  5.364  2.898
s(days_since_tillage):as.factor(management)organic   6.511e+00  6.895 39.707
                                                      p-value    
s(year):as.factor(management)conventional             < 2e-16 ***
s(year):as.factor(management)organic                  < 2e-16 ***
s(doy)                                                < 2e-16 ***
ti(soilt_avg,vwc_avg):as.factor(management)conventional  < 2e-16 ***
ti(soilt_avg,vwc_avg):as.factor(management)organic   6.55e-06 ***
s(soilt_avg):as.factor(management)conventional        < 2e-16 ***
```

```
s(soilt_avg):as.factor(management)organic                < 2e-16 ***
s(VPD):as.factor(management)conventional                 0.03793 *
s(VPD):as.factor(management)organic                      < 2e-16 ***
s(vwc_avg):as.factor(management)conventional             < 2e-16 ***
s(vwc_avg):as.factor(management)organic                 1.62e-05 ***
s(days_since_tillage):as.factor(management)conventional  0.00919 **
s(days_since_tillage):as.factor(management)organic       < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.866   Deviance explained = 87.2%
-REML = 2713.1  Scale est. = 1.0751    n = 1788
```

Interpretation, linear variables:

Note: baseline comparison to conventional management and mature crop stage (estimated Reco = 3.07)

Organic management has a significant effect on Reco baseline (conventional, mature) overall of about 1 unit. The crop stage also has significant effects on Reco – vegetative, reproductive, and grain fill respectively. The interactions between crop stage and management is also significant, for grain fill and trending that way for dormant. Again, the interaction coefficient tells us how much more or less the difference between that crop stage, and the baseline (mature), changes under organic compared to conventional management

Interpretaton, smoothing variables:

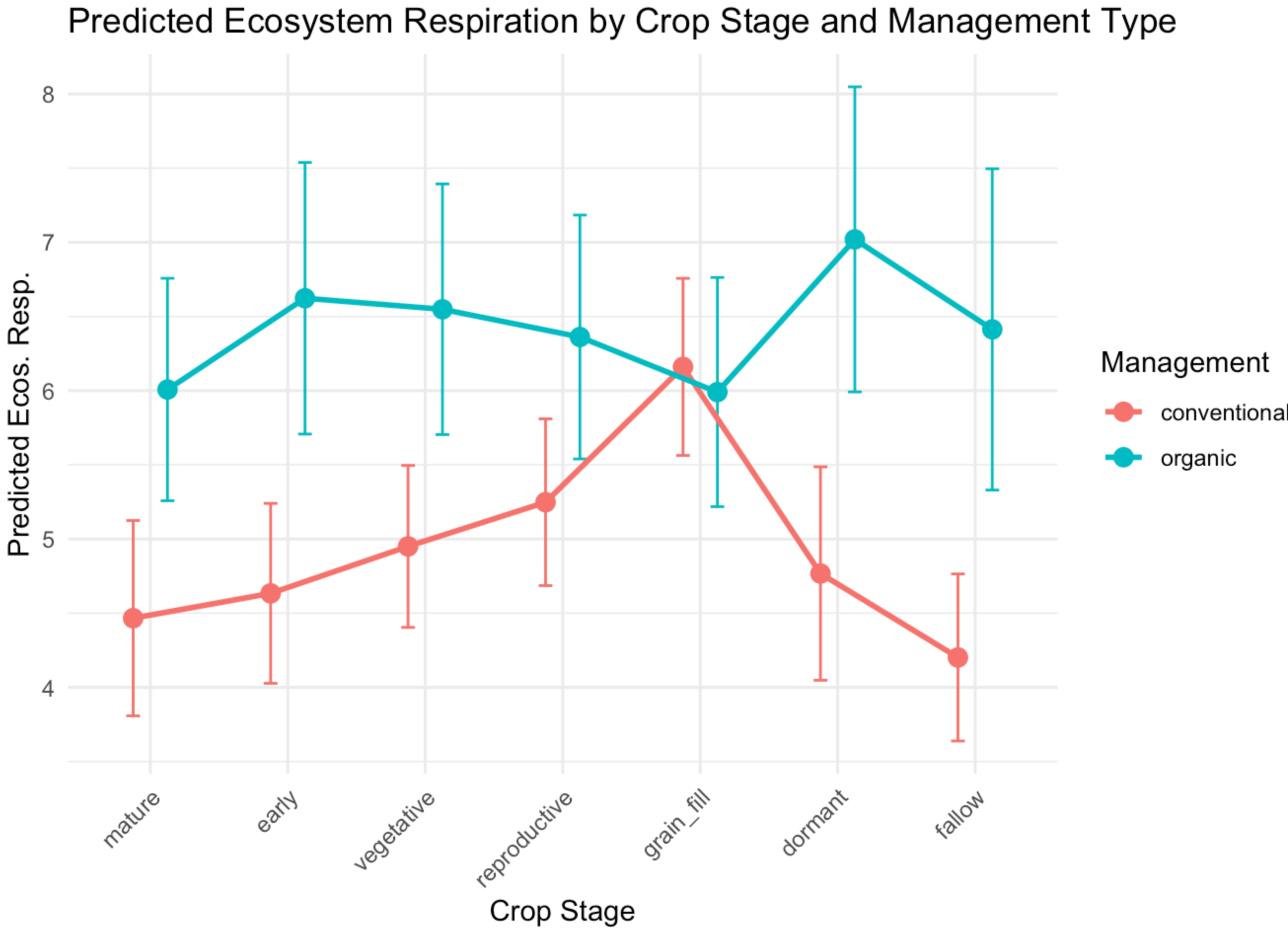
all smoothing variables are significant in their effects on Reco, and much like NEE, organic management tends toward more nonlinearity in each variable’s effects on Reco.

Takeaways:

Again and similar to NEE, organic management has a generally significant effect; in this case, in elevating ecosystem respiration across the board. Under conventional (baseline) management, the vegetative, reproductive, and grain fill stages also show significantly higher Reco, indicating that Reco tends to increase as crops develop and peaks at grain fill. In terms of interactions between management and crop stage, there is evidence that the increase in respiration at grain fill is dampened under organic management compared to the same stage under conventional management; so while respiration increases in organic generally, the ‘burst’ of respiration that occurs around grain fill is higher in the conventional field than in the organic one. Generally speaking, there are nonlinear and greater-in-complexity smoothing variables that represent edaphic characteristics in the organic field, indicating again greater complexity in the soil environment, microbiome, and/or the plant-soil interactions there.

Visualize:

Interactions, Reco daily average:



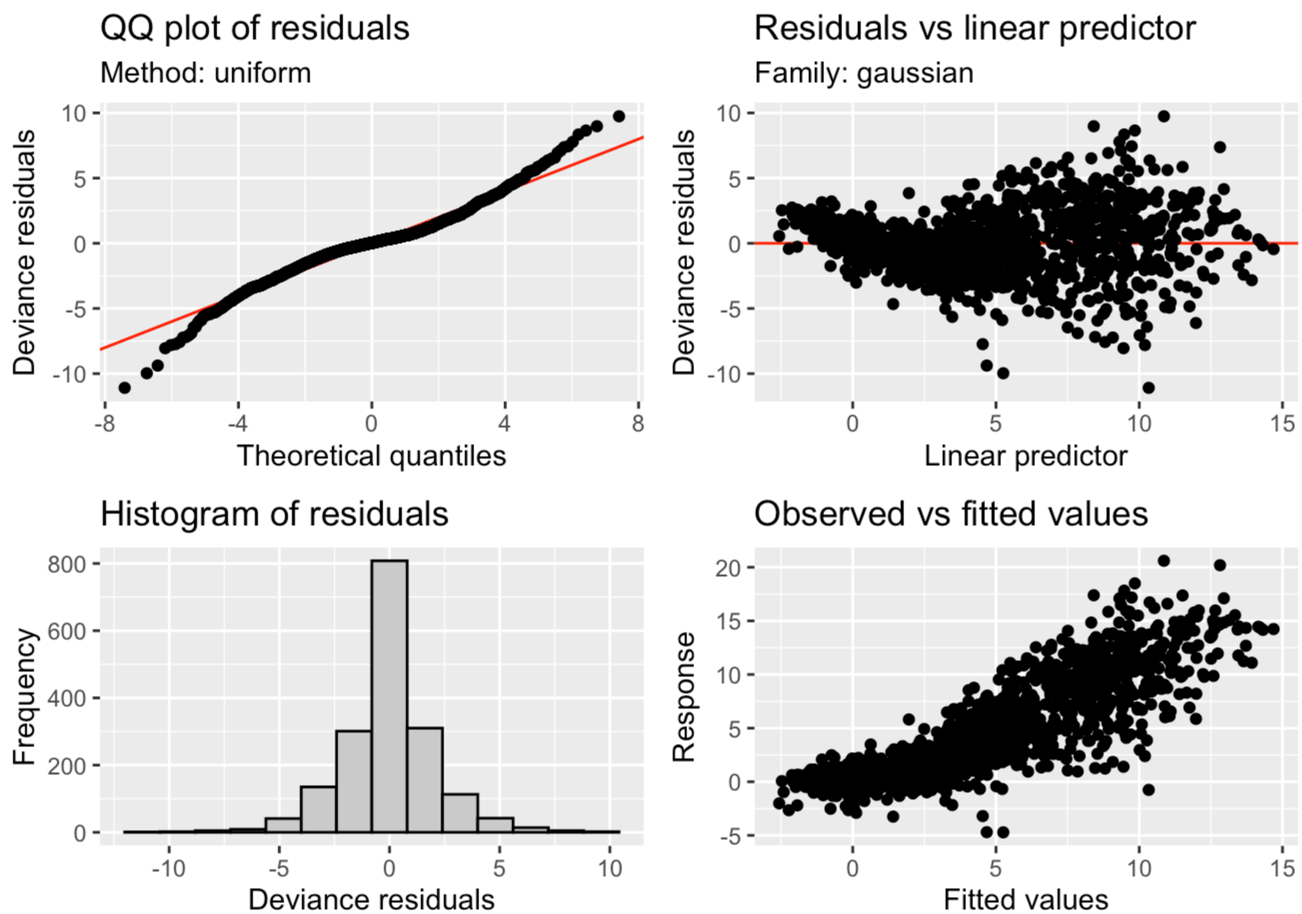
GPP

GAM model development: daily average:

```
           df      AIC
p1   64.28674 7953.106
p1.2 84.91324 7923.709
p1.3 91.95485 7912.924

Analysis of Deviance Table

Model 1: GPP_f ~ management + crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
Model 2: GPP_f ~ management * crop_stage_simple + s(year, bs = "re", by =
as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
```

```
    k = 8)
  Resid. Df Resid. Dev     Df Deviance      F   Pr(>F)
1    1689.6     8003.2
2    1682.4     7892.7 7.1935   110.54 3.3292 0.001412 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assess model performance:




```
Family: gaussian
Link function: identity

Formula:
GPP_f ~ management * crop_stage_simple + s(year, bs = "re", by = as.factor(management)) +
    s(doy, bs = "cc", k = 20) + ti(soilt_avg, vwc_avg, by = as.factor(management),
    k = c(8, 8)) + s(soilt_avg, by = as.factor(management), k = 10) +
    s(VPD, by = as.factor(management), k = 10) + s(vwc_avg, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)

Parametric coefficients:
                                               Estimate Std. Error t value
(Intercept)                                     2.84717    0.38466   7.402
managementorganic                               1.81161    0.46523   3.894
crop_stage_simpleearly                         -0.91447    0.47810  -1.913
```

```
crop_stage_simplevegetative                      0.09474    0.37670   0.252
crop_stage_simplereproductive                    0.42961    0.42181   1.018
crop_stage_simplegrain_fill                      1.82900    0.38676   4.729
crop_stage_simpledormant                         0.51022    0.51014   1.000
crop_stage_simplefallow                         -1.80087    0.40051  -4.496
managementorganic:crop_stage_simpleearly        -0.66745    0.65372  -1.021
managementorganic:crop_stage_simplevegetative   -0.32563    0.50981  -0.639
managementorganic:crop_stage_simplereproductive  0.43580    0.57216   0.762
managementorganic:crop_stage_simplegrain_fill   -0.45068    0.50745  -0.888
managementorganic:crop_stage_simpledormant       1.60953    0.71505   2.251
managementorganic:crop_stage_simplefallow       -0.68404    0.84128  -0.813
                                                Pr(>|t|)
(Intercept)                                     2.10e-13 ***
managementorganic                               0.000102 ***
crop_stage_simpleearly                          0.055952 .
crop_stage_simplevegetative                     0.801456
crop_stage_simplereproductive                   0.308592
crop_stage_simplegrain_fill                     2.44e-06 ***
crop_stage_simpledormant                        0.317368
crop_stage_simplefallow                         7.38e-06 ***
managementorganic:crop_stage_simpleearly        0.307401
managementorganic:crop_stage_simplevegetative   0.523089
managementorganic:crop_stage_simplereproductive 0.446360
managementorganic:crop_stage_simplegrain_fill   0.374596
managementorganic:crop_stage_simpledormant      0.024516 *
managementorganic:crop_stage_simplefallow       0.416276
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                                              edf Ref.df      F
s(year):as.factor(management)conventional               9.840e-08  1.000  0.000
s(year):as.factor(management)organic                    2.842e-08  1.000  0.000
s(doy)                                                  1.409e+01 18.000 15.607
ti(soilt_avg,vwc_avg):as.factor(management)conventional 6.504e+00  9.025  3.697
ti(soilt_avg,vwc_avg):as.factor(management)organic      8.020e+00 11.402  1.516
s(soilt_avg):as.factor(management)conventional          4.397e+00  5.354 13.748
s(soilt_avg):as.factor(management)organic               1.016e+00  1.028  0.650
s(VPD):as.factor(management)conventional                3.472e+00  4.385  7.010
s(VPD):as.factor(management)organic                     5.145e+00  6.251 30.145
s(vwc_avg):as.factor(management)conventional            4.584e+00  5.684  2.198
s(vwc_avg):as.factor(management)organic                 6.176e+00  7.332  2.886
s(days_since_tillage):as.factor(management)conventional 4.313e+00  5.170  3.054
s(days_since_tillage):as.factor(management)organic      6.280e+00  6.794 11.909
                                                         p-value
s(year):as.factor(management)conventional                < 2e-16 ***
s(year):as.factor(management)organic                    0.022934 *
s(doy)                                                   < 2e-16 ***
ti(soilt_avg,vwc_avg):as.factor(management)conventional 0.000138 ***
ti(soilt_avg,vwc_avg):as.factor(management)organic      0.117302
s(soilt_avg):as.factor(management)conventional           < 2e-16 ***
```

```
s(soilt_avg):as.factor(management)organic             0.425554
s(VPD):as.factor(management)conventional              9.34e-06 ***
s(VPD):as.factor(management)organic                    < 2e-16 ***
s(vwc_avg):as.factor(management)conventional          0.033029 *
s(vwc_avg):as.factor(management)organic               0.005219 **
s(days_since_tillage):as.factor(management)conventional 0.009417 **
s(days_since_tillage):as.factor(management)organic     < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.761   Deviance explained = 77.1%
-REML =   3983  Scale est. = 4.6156    n = 1788
```

Interpretation, linear variables:

Baseline GPP (mature crop stage, conventional management) estimate: 2.85

Organic management increases GPP by about 1.8 units. Under conventional management, there is a trend toward lower GPP in the early phase compared to mature (makes sense) and significantly higher GPP in the crop fill stage (also makes sense). There is significantly lower GPP in the fallow phase (also expected). Like respiration, GPP peaks in the grain fill stage. Most interaction terms aren't significant, except that between organic management and dormat crop stage, which has a 1.6 unit increase compared to baseline, indicating that GPP is higher in the organic field during dormancy than in the conventional field during dormancy.

Interpretation, smoothing variables:

Unlike NEE and Reco, the tensor interaction between soil temp and moisture is significant only for the conventional field, indicating that there is a strong combined influence there but not the organic field. To that end, soil temperature is significant and very complex in the conventional field but not organic, while soil moisture is significant and complex in both fields but more so in the organic field. VPD is significant and nonlinear in the organic field as well, and days since tillage is again significant in both fields but more complex in the organic field.

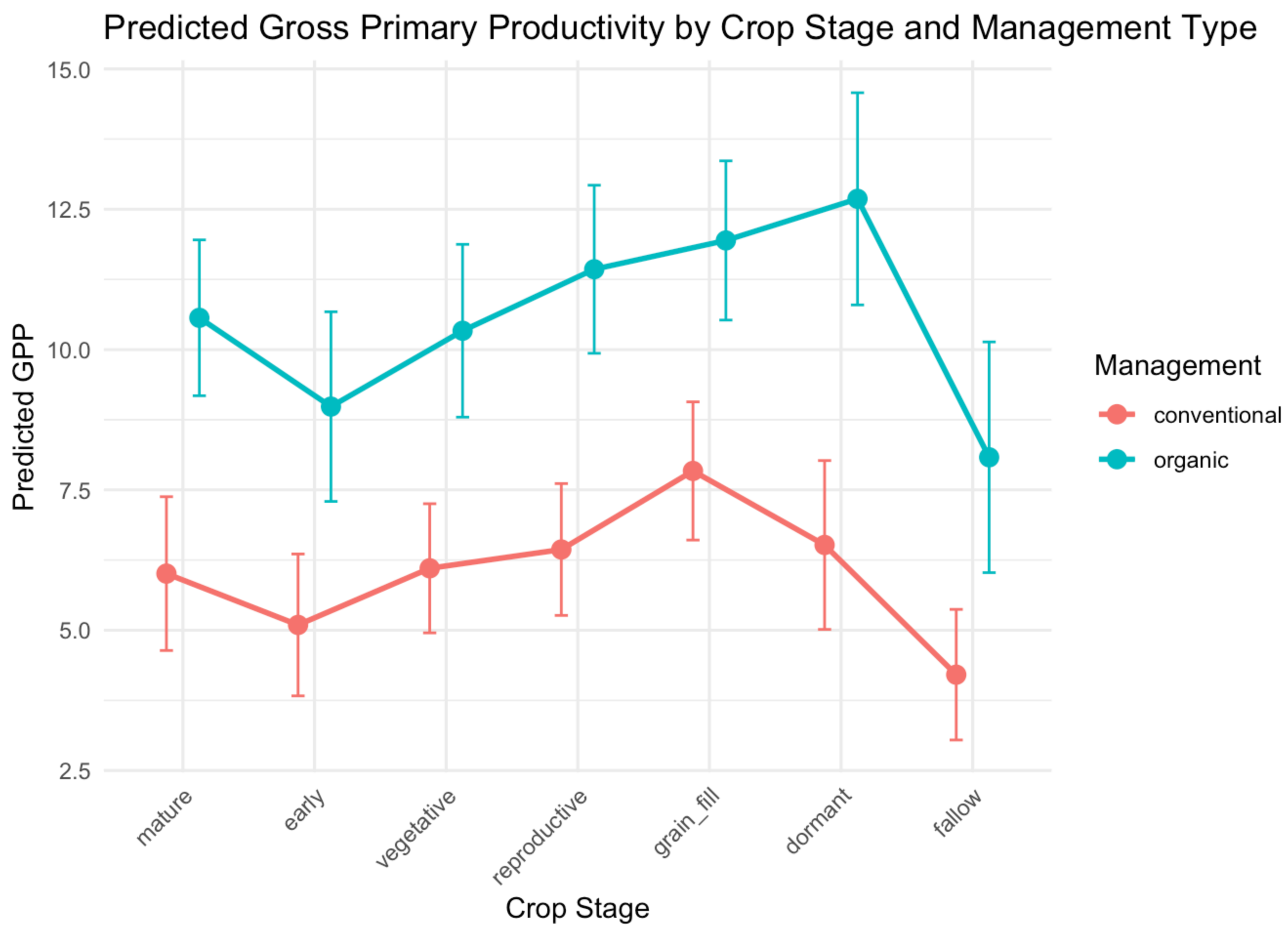
Takeaways:

As for NEE and Reco, organic management is highly significant and results in higher baseline GPP (and thus productivity). GPP is strongly moderated by crop stage, particularly grain fill (peak productivity) and dormant (low). An interesting note is that there is no significant interaction between crop stage and management except for dormancy in the organic field, when dormant cover crops over the winter months have greater GPP than those dormant cover crops in the conventional field.

The conventional field has strong nonlinear effects of soil temperature as well as the interaction between soil temperature and moisture, while the organic field does not; however, the organic field demonstrates more complex effects of the other edaphic drivers on GPP including soil moisture. As predicted and expected, seasonality and year-to-year variation are both significant drivers of GPP in all cases.

Visualize:

Interactions, GPP daily average:



GAM: chamber data, daily averages

NEE: daily average

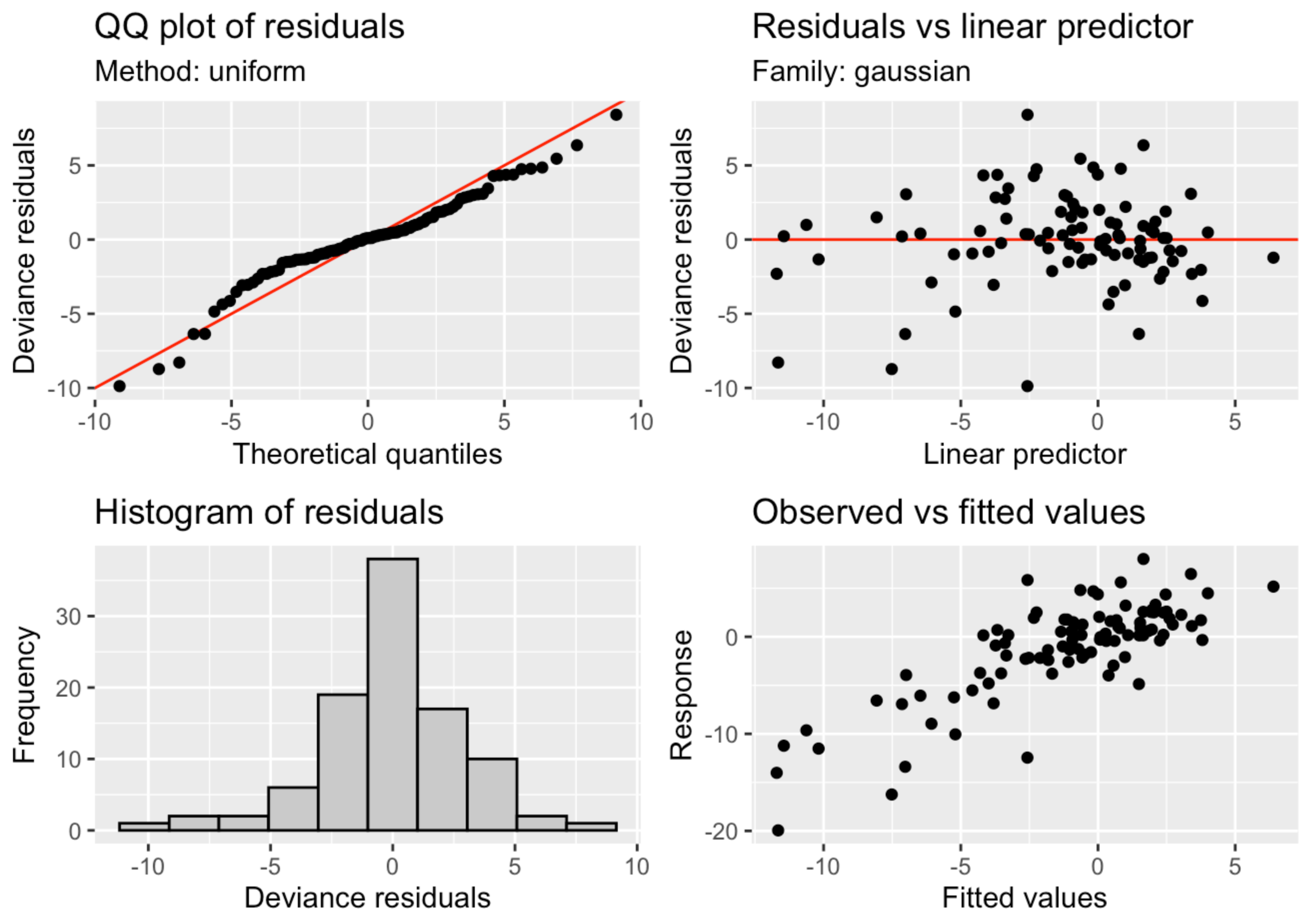
There are significantly fewer observations with the survey chambers, but in general we want to see agreement with the trends we saw in the eddy covariance dataset (at least for the years for which we have concurrent data: 2019 and 2020). Our daily averages are going to be based on two daytime observations, rather than 24hr averages. Thus the NEE and Rs values will be daytime only and perhaps a bit easier to interpret.

Model development:

```
         df      AIC
c1 18.16276 571.5739
c2 24.89599 567.2237
c3 32.87916 558.5610
```

```
Analysis of Deviance Table

Model 1: NEE_umolm2sec_mean ~ management + crop_stage_simple + s(year,
```

```
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(airtempC_mean, by = as.factor(management),
    k = 10) + s(soilm_perc_mean, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
Model 2: NEE_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(airtempC_mean, by = as.factor(management),
    k = 10) + s(soilm_perc_mean, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
  Resid. Df Resid. Dev    Df Deviance      F  Pr(>F)
1    79.459     1351.3
2    72.437     1126.7 7.022   224.63 2.1514 0.04853 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Analysis of Deviance Table

Model 1: NEE_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(airtempC_mean, by = as.factor(management),
    k = 10) + s(soilm_perc_mean, by = as.factor(management),
    k = 10) + s(days_since_tillage, by = as.factor(management),
    k = 8)
Model 2: NEE_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + ti(soilm_perc_mean,
    airtempC_mean, by = as.factor(management), k = c(8, 8)) +
    s(airtempC_mean, by = as.factor(management), k = 10) + s(soilm_perc_mean,
    by = as.factor(management), k = 10) + s(days_since_tillage,
    by = as.factor(management), k = 8)
  Resid. Df Resid. Dev     Df Deviance      F  Pr(>F)
1    72.437    1126.65
2    62.424     876.28 10.013   250.37 1.9923 0.04917 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```




```
Family: gaussian
Link function: identity

Formula:
NEE_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + ti(soilm_perc_mean,
    airtempC_mean, by = as.factor(management), k = c(8, 8)) +
    s(airtempC_mean, by = as.factor(management), k = 10) + s(soilm_perc_mean,
    by = as.factor(management), k = 10) + s(days_since_tillage,
    by = as.factor(management), k = 8)

Parametric coefficients:
                                            Estimate Std. Error t value
(Intercept)                                    5.943      3.728   1.594
managementorganic                            -12.531     10.474  -1.196
crop_stage_simpleearly                        -5.865      3.549  -1.652
crop_stage_simplevegetative                   -5.092      3.458  -1.473
crop_stage_simplereproductive                 -9.129      3.589  -2.543
crop_stage_simplegrain_fill                   -9.982      3.318  -3.009
crop_stage_simplefallow                       -3.935      3.734  -1.054
managementorganic:crop_stage_simpleearly       8.739      4.202   2.080
managementorganic:crop_stage_simplevegetative 11.699      4.180   2.799
managementorganic:crop_stage_simplereproductive 13.283    4.452   2.983
```

```
managementorganic:crop_stage_simplegrain_fill      9.498      4.098   2.318
managementorganic:crop_stage_simplefallow          0.000      0.000     NaN
                                                 Pr(>|t|)
(Intercept)                                       0.11537
managementorganic                                 0.23558
crop_stage_simpleearly                            0.10294
crop_stage_simplevegetative                       0.14534
crop_stage_simplereproductive                     0.01320 *
crop_stage_simplegrain_fill                       0.00365 **
crop_stage_simplefallow                           0.29571
managementorganic:crop_stage_simpleearly          0.04122 *
managementorganic:crop_stage_simplevegetative     0.00662 **
managementorganic:crop_stage_simplereproductive  0.00393 **
managementorganic:crop_stage_simplegrain_fill     0.02339 *
managementorganic:crop_stage_simplefallow             NaN
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                                                          edf
s(year):as.factor(management)conventional                           1.470e-08
s(year):as.factor(management)organic                                2.026e-07
s(month):as.factor(management)conventional                          6.462e-05
s(month):as.factor(management)organic                               3.359e-01
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional 5.411e+00
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic      1.005e+00
s(airtempC_mean):as.factor(management)conventional                  1.000e+00
s(airtempC_mean):as.factor(management)organic                       1.000e+00
s(soilm_perc_mean):as.factor(management)conventional                1.170e+00
s(soilm_perc_mean):as.factor(management)organic                     1.710e+00
s(days_since_tillage):as.factor(management)conventional             1.635e+00
s(days_since_tillage):as.factor(management)organic                  3.916e+00
                                                                    Ref.df
s(year):as.factor(management)conventional                            1.000
s(year):as.factor(management)organic                                 1.000
s(month):as.factor(management)conventional                           4.000
s(month):as.factor(management)organic                                4.000
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional  7.428
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic       1.009
s(airtempC_mean):as.factor(management)conventional                   1.000
s(airtempC_mean):as.factor(management)organic                        1.000
s(soilm_perc_mean):as.factor(management)conventional                 1.300
s(soilm_perc_mean):as.factor(management)organic                      2.171
s(days_since_tillage):as.factor(management)conventional              1.965
s(days_since_tillage):as.factor(management)organic                   4.387
                                                                        F
s(year):as.factor(management)conventional                           0.000
s(year):as.factor(management)organic                                0.000
s(month):as.factor(management)conventional                          0.000
s(month):as.factor(management)organic                               0.101
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional 1.435
```

```
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic      0.040
s(airtempC_mean):as.factor(management)conventional                  0.069
s(airtempC_mean):as.factor(management)organic                       1.126
s(soilm_perc_mean):as.factor(management)conventional                0.634
s(soilm_perc_mean):as.factor(management)organic                     1.229
s(days_since_tillage):as.factor(management)conventional             3.683
s(days_since_tillage):as.factor(management)organic                  3.260
                                                                    p-value
s(year):as.factor(management)conventional                            0.5000
s(year):as.factor(management)organic                                 0.5000
s(month):as.factor(management)conventional                           0.9778
s(month):as.factor(management)organic                                0.5597
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional  0.3481
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic       0.8554
s(airtempC_mean):as.factor(management)conventional                   0.7938
s(airtempC_mean):as.factor(management)organic                        0.2922
s(soilm_perc_mean):as.factor(management)conventional                 0.5488
s(soilm_perc_mean):as.factor(management)organic                      0.2762
s(days_since_tillage):as.factor(management)conventional              0.0350 *
s(days_since_tillage):as.factor(management)organic                   0.0139 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Rank: 169/170
R-sq.(adj) =  0.499   Deviance explained = 63.9%
-REML =  238.4  Scale est. = 12.551    n = 98
```

Interpretation:

While there is no significant effect of management on chamber-based daily average NEE, there it is clear that both crop stage (reproductive, grain fill) both reduce NEE when compared to the baseline of conventional, mature crop by quite a bit. Additionally, the interaction between management and crop stage is significant, with enhanced effects of early, vegetative, reproductive, and grain fill stages on NEE in the organic field compared to the conventional one.

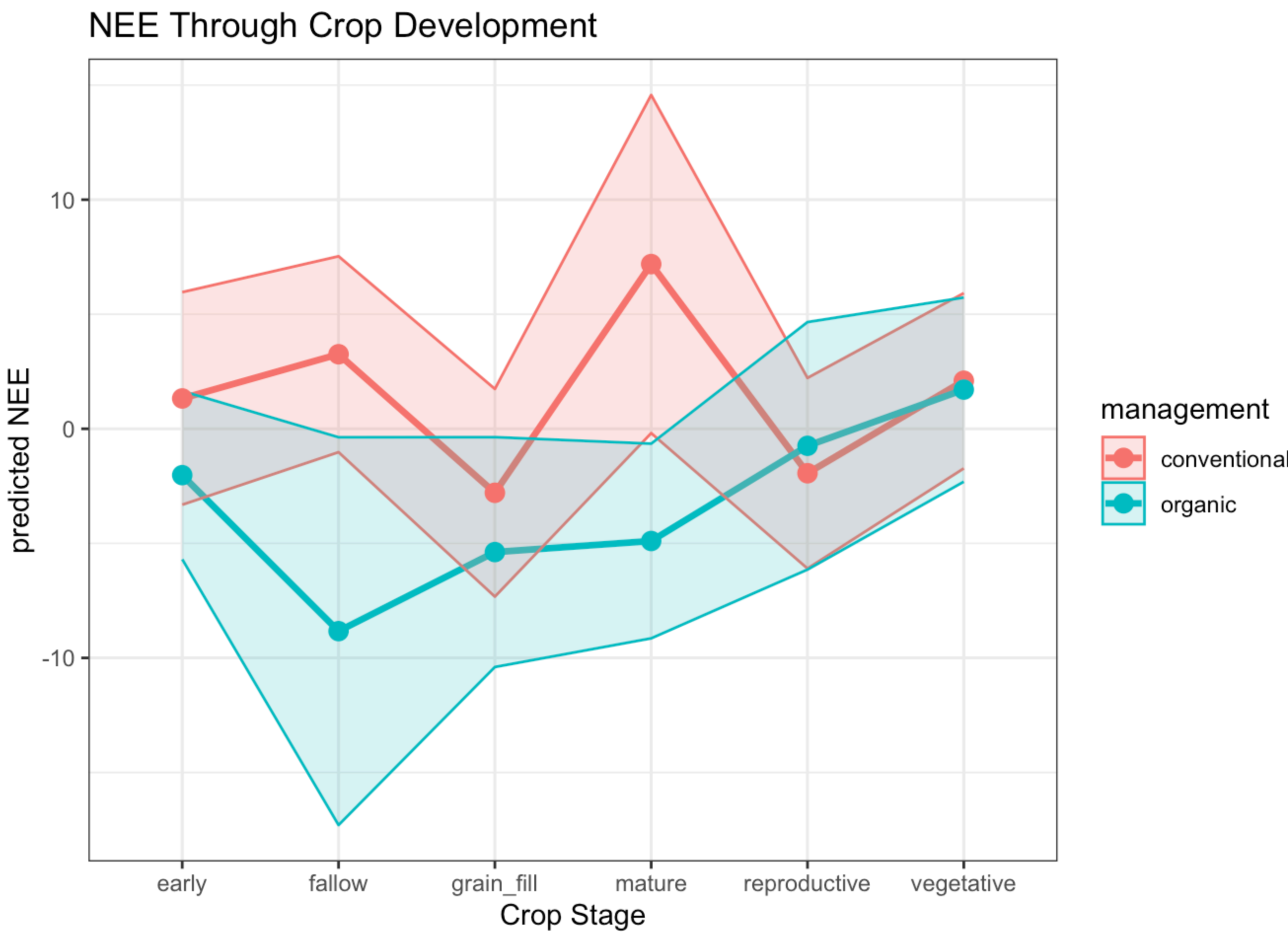
The only smoothing factors that are significant in driving NEE are days since tillage, though the effect is much more complex in the organic field than the conventional one.

Takeaways:

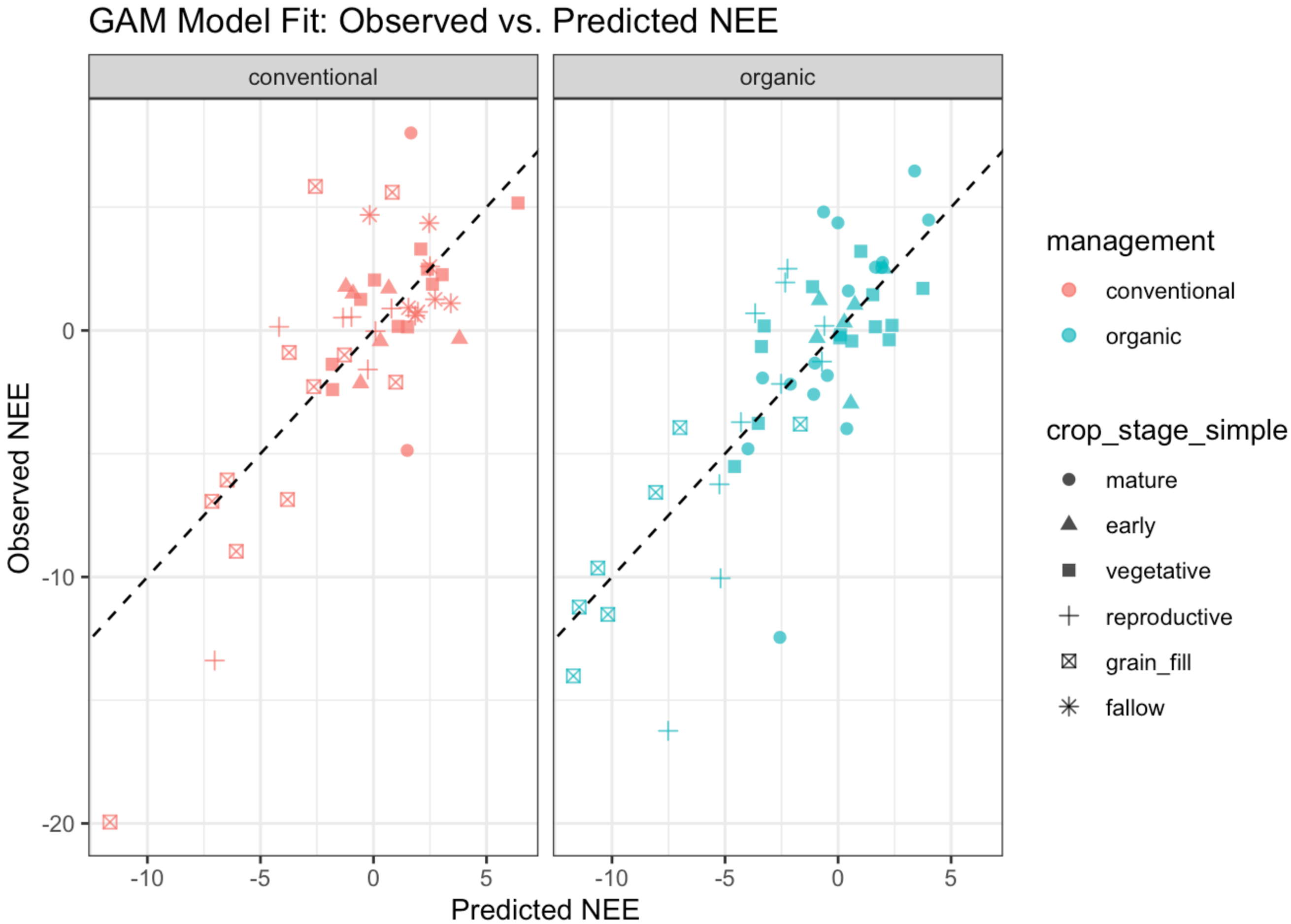
These results largely align with those data collected from the EC towers, reinforcing the conclusions that organic management generally reduces NEE but that this effect varies by crop stage. The drivers of the effect are much less linear / more complex in the organic field compared to the conventional one.

Visualize:

Predicted Through Crop Stage:



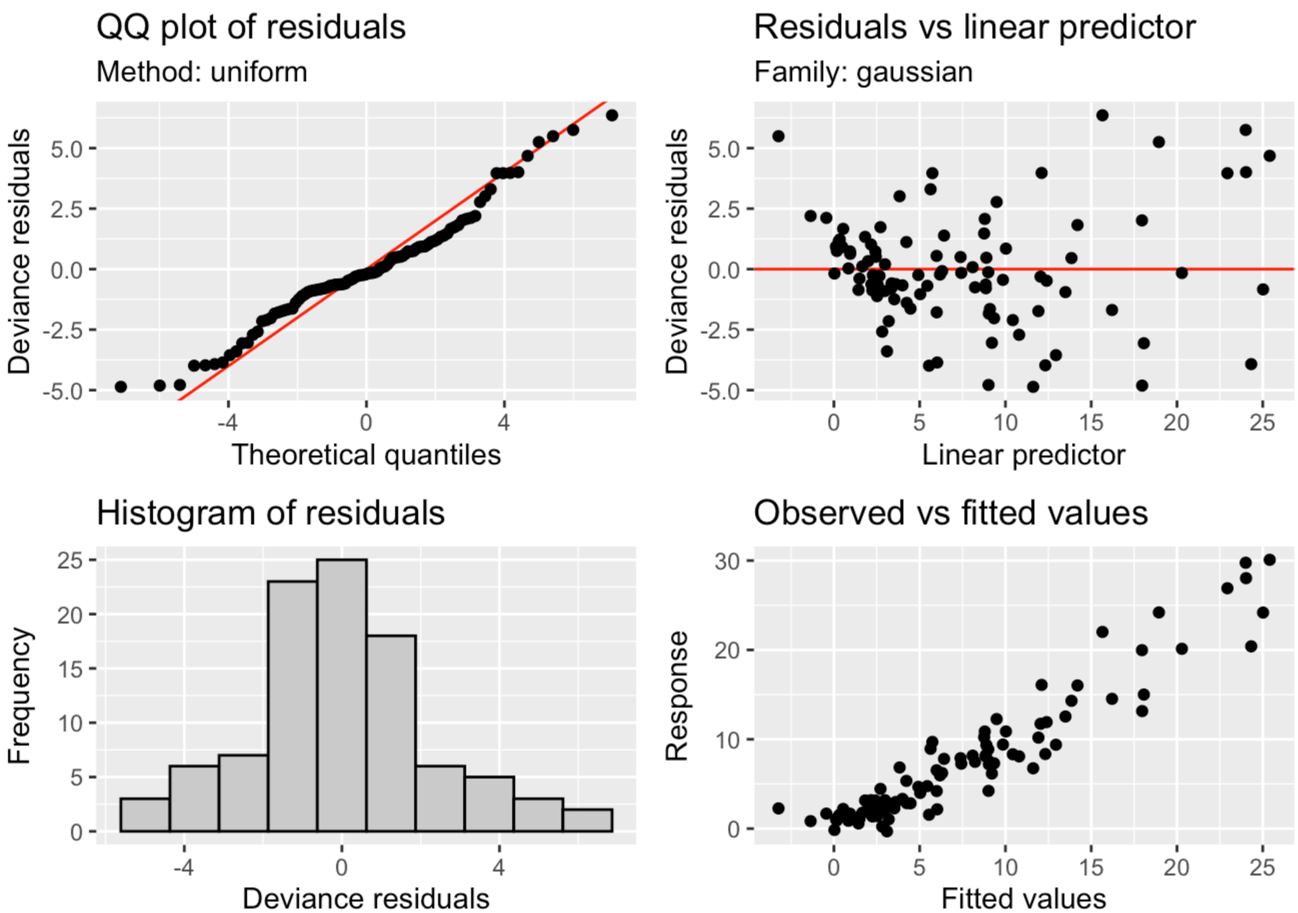
Predicted vs. Observed



Reco: daily average

```
          df      AIC
n1 25.25305 538.8713
n2 29.72541 507.1832
n3 33.21324 508.7876

Analysis of Deviance Table

Model 1: Rs_umolm2sec_mean ~ management + crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(soilm_perc_mean, k = 10,
    by = as.factor(management)) + s(airtempC_mean, k = 10, by = as.factor(management)) +
    s(days_since_tillage, k = 8, by = as.factor(management))
Model 2: Rs_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(soilm_perc_mean, k = 10,
    by = as.factor(management)) + s(airtempC_mean, k = 10, by = as.factor(management)) +
    s(days_since_tillage, k = 8, by = as.factor(management))
  Resid. Df Resid. Dev     Df Deviance      F    Pr(>F)
1    72.239     837.49
2    67.726     553.24 4.5126   284.25 8.0638 1.034e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Analysis of Deviance Table

Model 1: Rs_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + s(soilm_perc_mean, k = 10,
    by = as.factor(management)) + s(airtempC_mean, k = 10, by = as.factor(management)) +
    s(days_since_tillage, k = 8, by = as.factor(management))
Model 2: Rs_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + ti(soilm_perc_mean,
    airtempC_mean, by = as.factor(management), k = c(8, 8)) +
    s(soilm_perc_mean, k = 10, by = as.factor(management)) +
    s(airtempC_mean, k = 10, by = as.factor(management)) + s(days_since_tillage,
    k = 8, by = as.factor(management))
  Resid. Df Resid. Dev     Df Deviance     F Pr(>F)
1    67.726     553.24
2    63.401     523.73 4.3252   29.506 0.888 0.4828
```




```
Family: gaussian
Link function: identity

Formula:
Rs_umolm2sec_mean ~ management * crop_stage_simple + s(year,
    bs = "re", by = as.factor(management)) + s(month, bs = "cc",
    k = 6, by = as.factor(management)) + ti(soilm_perc_mean,
```

```
    airtempC_mean, by = as.factor(management), k = c(8, 8)) +
    s(soilm_perc_mean, k = 10, by = as.factor(management)) +
    s(airtempC_mean, k = 10, by = as.factor(management)) + s(days_since_tillage,
    k = 8, by = as.factor(management))

Parametric coefficients:
                                                 Estimate Std. Error t value
(Intercept)                                       4.33021    2.76317   1.567
managementorganic                                 1.16566   13.14109   0.089
crop_stage_simpleearly                           -4.25172    2.65227  -1.603
crop_stage_simplevegetative                      -2.16520    2.53564  -0.854
crop_stage_simplereproductive                     4.73589    2.74388   1.726
crop_stage_simplegrain_fill                       7.79382    2.63126   2.962
crop_stage_simplefallow                          -0.05764    2.85150  -0.020
managementorganic:crop_stage_simpleearly          1.68518    3.21576   0.524
managementorganic:crop_stage_simplevegetative    -2.59409    3.19680  -0.811
managementorganic:crop_stage_simplereproductive -11.88158    3.56474  -3.333
managementorganic:crop_stage_simplegrain_fill   -13.23215    3.33267  -3.970
managementorganic:crop_stage_simplefallow         0.00000    0.00000     NaN
                                                Pr(>|t|)
(Intercept)                                     0.121718
managementorganic                               0.929578
crop_stage_simpleearly                          0.113548
crop_stage_simplevegetative                     0.396148
crop_stage_simplereproductive                   0.088881 .
crop_stage_simplegrain_fill                     0.004204 **
crop_stage_simplefallow                         0.983933
managementorganic:crop_stage_simpleearly        0.601952
managementorganic:crop_stage_simplevegetative   0.419925
managementorganic:crop_stage_simplereproductive 0.001391 **
managementorganic:crop_stage_simplegrain_fill   0.000175 ***
managementorganic:crop_stage_simplefallow            NaN
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                                                        edf
s(year):as.factor(management)conventional                         1.047e-08
s(year):as.factor(management)organic                              1.712e-07
s(month):as.factor(management)conventional                        3.458e+00
s(month):as.factor(management)organic                             2.608e+00
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional 1.164e+00
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic      1.772e+00
s(soilm_perc_mean):as.factor(management)conventional              1.000e+00
s(soilm_perc_mean):as.factor(management)organic                   1.000e+00
s(airtempC_mean):as.factor(management)conventional                1.000e+00
s(airtempC_mean):as.factor(management)organic                     1.000e+00
s(days_since_tillage):as.factor(management)conventional           1.372e+00
s(days_since_tillage):as.factor(management)organic                4.453e+00
                                                                    Ref.df
s(year):as.factor(management)conventional                            1.000
```

```
s(year):as.factor(management)organic                                  1.000
s(month):as.factor(management)conventional                            4.000
s(month):as.factor(management)organic                                 4.000
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional  1.318
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic       2.221
s(soilm_perc_mean):as.factor(management)conventional                  1.000
s(soilm_perc_mean):as.factor(management)organic                       1.000
s(airtempC_mean):as.factor(management)conventional                    1.000
s(airtempC_mean):as.factor(management)organic                         1.000
s(days_since_tillage):as.factor(management)conventional               1.628
s(days_since_tillage):as.factor(management)organic                    4.815
                                                                          F
s(year):as.factor(management)conventional                             0.000
s(year):as.factor(management)organic                                  0.000
s(month):as.factor(management)conventional                           13.392
s(month):as.factor(management)organic                                 2.714
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional  1.611
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic       1.251
s(soilm_perc_mean):as.factor(management)conventional                  0.738
s(soilm_perc_mean):as.factor(management)organic                       1.334
s(airtempC_mean):as.factor(management)conventional                    0.076
s(airtempC_mean):as.factor(management)organic                         4.372
s(days_since_tillage):as.factor(management)conventional              12.536
s(days_since_tillage):as.factor(management)organic                   13.206
                                                                      p-value
s(year):as.factor(management)conventional                              0.5000
s(year):as.factor(management)organic                                   0.9995
s(month):as.factor(management)conventional                             0.3743
s(month):as.factor(management)organic                                  0.0735
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional   0.2612
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic        0.2881
s(soilm_perc_mean):as.factor(management)conventional                   0.3932
s(soilm_perc_mean):as.factor(management)organic                        0.2521
s(airtempC_mean):as.factor(management)conventional                     0.7835
s(airtempC_mean):as.factor(management)organic                          0.0403
s(days_since_tillage):as.factor(management)conventional              9.18e-05
s(days_since_tillage):as.factor(management)organic                   3.15e-06

s(year):as.factor(management)conventional
s(year):as.factor(management)organic
s(month):as.factor(management)conventional
s(month):as.factor(management)organic                                .
ti(soilm_perc_mean,airtempC_mean):as.factor(management)conventional
ti(soilm_perc_mean,airtempC_mean):as.factor(management)organic
s(soilm_perc_mean):as.factor(management)conventional
s(soilm_perc_mean):as.factor(management)organic
s(airtempC_mean):as.factor(management)conventional
s(airtempC_mean):as.factor(management)organic                        *
s(days_since_tillage):as.factor(management)conventional              ***
s(days_since_tillage):as.factor(management)organic                   ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Rank: 169/170
R-sq.(adj) =   0.85   Deviance explained = 89.5%
-REML = 224.97  Scale est. = 7.6824    n = 98
```

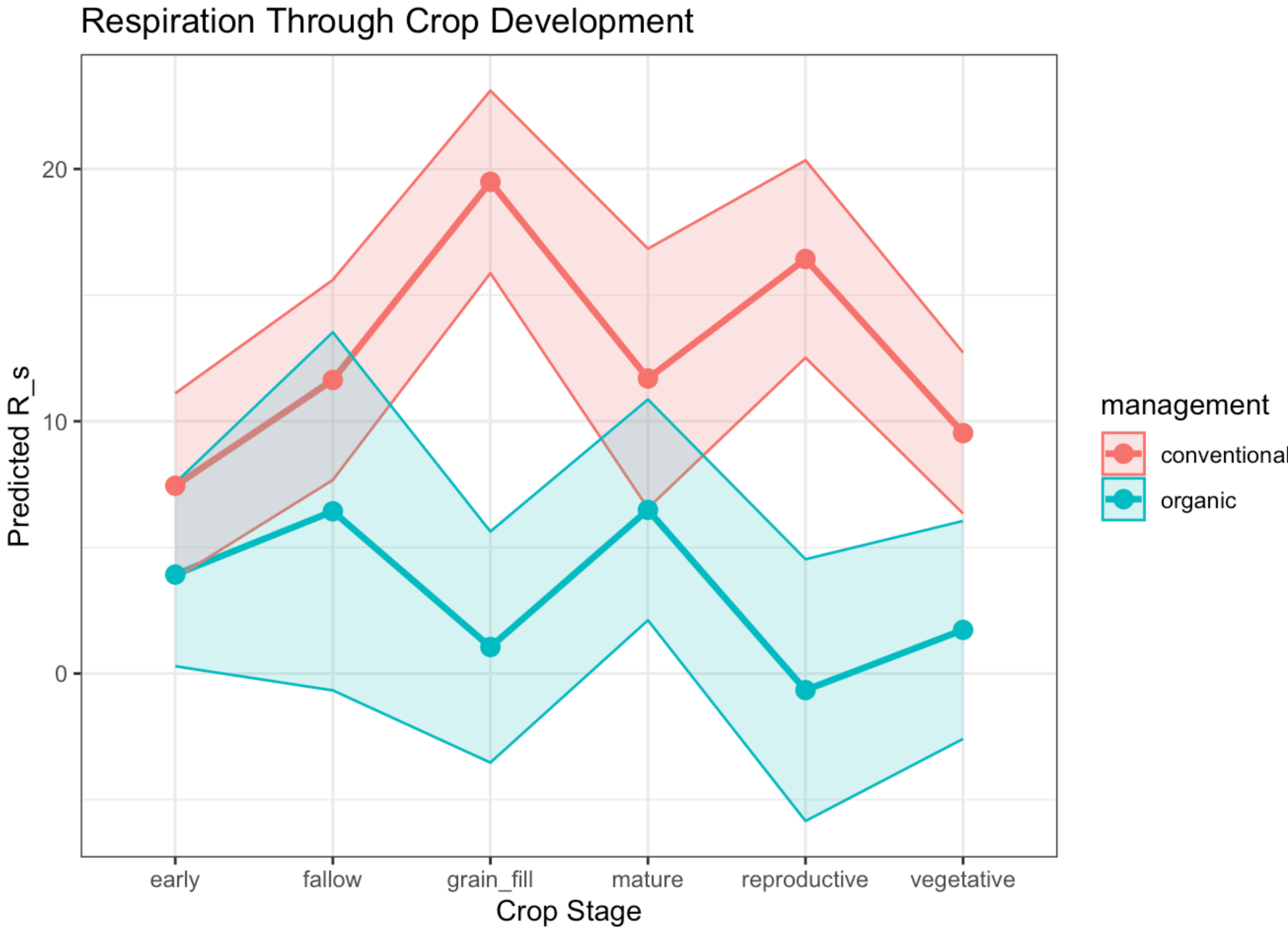
#####Interpretation: Again, management does not have a clear significant effect when holding all else equal. However, crop stage once again indicates differences in ecosystem respiration with fallow having significantly higher respiration when compared to conventional mature, and grain fill indicating higher respiration as well. The interaction between management and crop stage is significant for the reproductive and grain fill stages, indicating a much lower respiration rate in these stages when compared to conventional mature.

As for chamber-based NEE, chamber-based ecosystem respiraiton is significantly moderated by days since tillage and once again, the effect is far more complex for the organic field than the conventional field. In addition, the organic field has a significant effect of air temperature, which is fairly linear in its positive effect, and seasonality (here, month) indicates a slightly positive effect in the organic field as well.

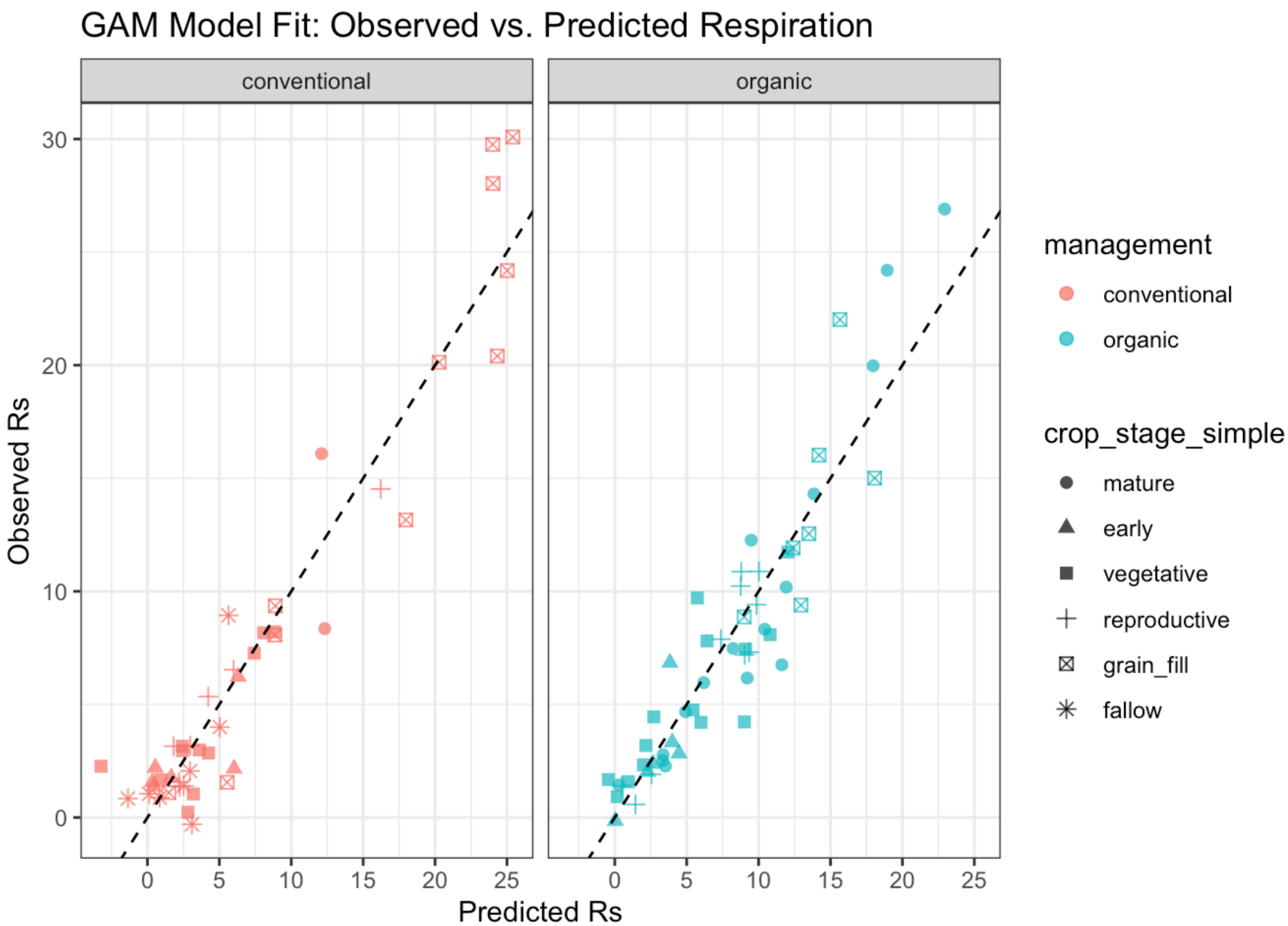
#####Takeaways: Again this analysis largely supports the conclusions from the EC tower data, and days since tillage seems to be the most prominent smoothing variable in both fields that emerges from the “all smooths are significant” outcomes of the tower data, in terms of moderating both NEE and ecosystem respiration on the scale of the collars.

Visualize:

Predicted Through Crop Stage:



Predicted vs. Observed


GAM Model Fit: Observed vs. Predicted Respiration
conventional
organic
Observed Rs
Predicted Rs
0
10
20
30
0
5
10
15
20
25
management
conventional
organic
crop_stage_simple
mature
early
vegetative
reproductive
grain_fill
fallow